Explicit two-sided vertex-minor universality from finite projective-plane incidence graphs

Maxime Cautres, Nathan Claudet, Mehdi Mhalla, Simon Perdrix, Valentin Savin, Stepha

arxiv:2402.06260v3; CC BY 4.0

Editorial note. Only theo:bipartite-kvmu is the counted target. Full author mathematical body, construction support, diagrams, bibliography and appendix retained. Publisher metadata/class is adapted to article; original source remains preserved.

Editorial note. Two native TikZ drawing comments missing the percent marker are marked as comments in assembly only; all figure nodes, edges and mathematical labels retained.

Allowed standard prerequisites (editorial).

- Standard algorithmic and mathematical background cited by the authors. All source-local constructions, proofs, appendices and references are retained; no new mathematical assumptions added.

Abstract

We study the notion of k-stabilizer universal quantum state, that is, an n-qubit quantum state, such that it is possible to induce any stabilizer state on any k qubits, by using only local operations and classical communications. These states generalize the notion of k-pairable states introduced by Bravyi et al., and can be studied from a combinatorial perspective using graph states and k-vertex-minor universal graphs. First, we demonstrate the existence of k-stabilizer universal graph states that are optimal in size with $n = \Theta(k^2)$ qubits. We also provide parameters for which a random graph state on $\Theta(k^2)$ qubits is k-stabilizer universal with high probability. Our second contribution consists of two explicit constructions of k-stabilizer universal graph states on $n = O(k^4)$ qubits. Both rely upon the incidence graph of the projective plane over a finite field $\mathbb{F}_q$. This provides a major improvement over the previously known explicit construction of k-pairable graph states with $n = O(2^{3k})$, bringing forth a new and potentially powerful family of multipartite quantum resources.

1 Introduction

Quantum communication networks often rely on classical communication along with pre-shared entanglement. In this context, a highly pertinent problem is to explore which resource states enable a group of n parties, equipped with the capability of employing Local Operations and Classical Communication (LOCC),

to create entangled EPR pairs among any k pairs of qubits. It is only recently that Bravyi et al. addressed this fundamental question and provided both upper and lower bounds for what they called the k-pairability of quantum states, in terms of the number of parties and the number of qubits per party needed for a quantum state to be k-pairable [2]. Formally, an n-party state $|\psi\rangle$ is said to be k-pairable if, for every k disjoint pairs of parties $\{a_1, b_1\}, \ldots, \{a_k, b_k\}$, there exists a LOCC protocol that starts with $|\psi\rangle$ and ends up with a state where each of those k pairs of parties shares an EPR pair. Bravyi et al. studied n-party states in the case where each party holds m qubits, with m ranging from 1 to $\log(n)$. In the case where each party holds at least $m = 10$ qubits, they showed the existence of k-pairable states where k is of the order of $n/\text{polylog}(n)$, which is nearly optimal when m is constant. They also showed that if one allows a logarithmic number of qubits per party, then there exist k-pairable states with $k = n/2$. Moreover, before their work, numerous variations of this problem had surfaced in the literature, some in the context of entanglement routing [15, 25, 26], and some about problems that can be described as variants of k-pairability, for example (i) preparing resource states by clustering and merging [24], or starting from a particular network [6, 11], (ii) creating one EPR pair hidden from other parties [17], (iii) studying the complexity and providing algorithms for generating EPR pairs within a predetermined set [7, 8], (iv) estimating the cost of distributing a graph state in terms of EPR pairs [12, 23], and (v) transforming an arbitrary entangled state into another through a network where each party holds multiple qubits [4].

The notion of k-pairability that we focus on in the present paper relates to the scenario that is both the most natural and challenging [2], when each party possesses precisely one qubit, i.e., $m = 1$. Protocols with multi-qubit parties, require the use of quantum operations acting on two (or more) qubits, which are significantly harder to implement in all the known technologies. For instance in quantum optics, whose 'flying' qubits are well suited for pairability protocols, one-qubit operations are easy to perform using off-the-shelf standard devices, whereas two-qubit operations, like those required by the protocols using multi-qubit parties, can only be performed probabilistically with a non-negligible probability of failure [1,13,20,21]. Bravyi et al. provided some results in the setup where each party holds one single qubit, although arguably weaker than those obtained in the case where each party holds at least 10 qubits. Using Reed-Muller codes, they were able to construct a k-pairable state of size exponential in k, namely $n = 2^{3k}$, leaving the existence of a k-pairable states of size $n = poly(k)$ as an open problem. They also found a 2-pairable graph state of size 10 and proved that there exists no stabilizer state on less than 10 qubits that is 2-pairable using LOCC protocol based on Pauli measurements.

A natural generalization is to consider quantum states satisfying a stronger property: for some integer k, it is possible to induce any stabilizer state on any subset of k qubits, by means of LOCC protocols. We call these states k-stabilizer universal. Stabilizer states constitute a powerful resource for multipartite quantum protocols [16, 18, 22, 27], and can be described, up to local [1] unitaries, by the formalism of graph states: a subset of quantum states which are in one-to-one correspondence with (undirected, simple) graphs. $2k$-stabilizer

[1] As we consider one qubit per party, "local" is to be understood as "on each single qubit independently".

universality is a stronger notion than k-pairability: any $2k$-stabilizer universal state is k-pairable, as EPR pairs are stabilizer states.

Our contributions rely on the graph state formalism and the ability to characterize properties of quantum states using tools from graph theory. In particular, we reformulate k-pairability as a property of a graph (rather than a property of a quantum state), such that the graph state corresponding to a k-pairable graph, is k-pairable. Furthermore, we relate pairability to the standard notion of vertex-minor (a complete and up-to-date survey on vertex-minors can be found in [19]). A graph H is a vertex-minor of G if one can transform G into H by means of local complementations[2] and vertex deletions. If H is a vertex-minor of G then the graph state $|H\rangle$ can be obtained from $|G\rangle$ using only local Clifford operations, local Pauli measurements and classical communications. Dahlberg, Helsen, and Wehner proved that the converse is also true when H has no isolated vertices [9]. In [8], they proved that it is NP-complete[3] to decide whether a graph state can be transformed into a set of EPR pairs on specific qubits using using only local Clifford operations, local Pauli measurements and classical communications. In [7], they showed that it is also NP-complete to decide whether a graph state can be transformed into another one using using only local Clifford operations, local Pauli measurements and classical communications.

The graphical counterpart of k-stabilizer universal graph states are called k-vertex-minor universal graphs, introduced in [5]: a graph is k-vertex minor universal if it has any graph defined on any k of its vertices as a vertex minor. If a graph is k-vertex-minor universal then the corresponding graph state is k-stabilizer universal. Stabilizer universal states (and thus k-vertex-minor universal graphs) are useful in themselves beyond the fact that they imply pairability, as they can serve as a primitive for quantum protocols using multipartite entanglement. For instance, in [5], it is shown that stabilizer universal states constitute a resource to perform a robust pairability protocol, in the sense that it allows some known parties to be malicious, while ensuring the correctness of the protocol. Furthermore, the notion of stabilizer universality is stronger than the notion of pairability. Nevertheless, while previous work [5] establishes the existence of k-stabilizer universal graph states of size $n = O(k^4 \ln(k))$ and of k-pairable graph states of size $n = O(k^3 \ln^3(k))$, there are no known graph states that are k-pairable but not $2k$-stabilizer universal.

In this work, we provide both probabilistic and explicit constructions of k-stabilizer universal graph states resulting from k-vertex-minor universal graphs. While the results are interesting in themselves from a combinatorial perspective, they allow one to explicitly define quantum communication protocols: if a k-stabilizer universal graph state is prepared, and each qubit is sent to a different party, then, with the assumption that each party can perform local quantum operations and that they can share classical information, any stabilizer state on any k qubits can be generated. Note that this includes any set of disjoint EPR pairs on less than k qubits. The local operations to perform in order to induce a given subgraph state derive directly from the proofs of our results.

The main contributions of the paper are as follows. In the first part of the paper, we prove the existence of k-vertex-minor universal graphs of order $n = O(k^2)$, which is optimal according to [5]. We adopt a probabilistic approach,

[2]Local complementation on a vertex u consists in complementing the subgraph induced by its neighbors.

[3]Where the size of the input is the number of qubits of the graph state.

exhibiting a family of random bipartite graphs of quadratic order in k, which are k-vertex-minor universal with probability going to 1 exponentially fast in k. On the practical side, in the proof we introduce an efficient algorithm that tries to generate any induced graph of order k as a vertex-minor on any k vertices of a random bipartite graph, and the proof yields a bound on the probability of failure of the algorithm.

The second part of the paper focuses on explicit constructions of k-vertex-minor universal graphs. We derive our constructions from the incidence graph of the projective plane over the finite field $\mathbb{F}_q$, where q is a prime power. We denote this graph by $\mathbb{G}_q$. It is a bipartite graph of order $n = 2(q^2+q+1)$, with the same number $(n/2)$ of left and right vertices, corresponding respectively to points and lines of the projective plane (equivalently, 1-dimensional and 2-dimensional linear subspaces of $\mathbb{F}_q^3$). We show that $\mathbb{G}_q$ satisfies the k-vertex-minor universality property, with $k = \Theta(n^{1/4})$. Furthermore, we show that the graph on the points of the projective plane, with edges connecting points corresponding to orthogonal 1-dimensional linear subspaces of $\mathbb{F}_q^3$, is k-vertex-minor universal, again with $k = \Theta(n^{1/4})$. To the best of our knowledge, these are the first explicit constructions of k-vertex-minor universal graphs of order polynomial in k, significantly improving on the previous explicit construction of k-pairable states based on Reed-Muller codes from [2], with exponential overhead.

2 Vertex-minor and stabilizer universality

The goal of this section is to cover all notions gravitating around k-pairability and k-vertex-minor universality properties. We first define the above properties for graphs, then we discuss their implications on the corresponding graph states.

We denote a graph as $G = (V, E)$, where V is the vertex set and E is the edge set. All graphs are assumed to be undirected and simple (without loops or multiple edges). A vertex subset $S \subseteq V$ is said to be stable if no two vertices in S are adjacent. Bipartite graphs are denoted as $G = (L, R, E)$, where $V = L \sqcup R$, with L and R disjoint stable sets referred to as left and right vertex sets, respectively. To avoid possible confusion, we may sometimes write $V(G), E(G), L(G)$, or $R(G)$. A pairing is a graph G such that any vertex is incident to exactly one edge. Given a vertex $v \in V$ and a subset $U \subseteq V$, we denote by $N_G(v)$ the neighborhood of v in G, consisting of vertices $v \in V$ adjacent to v.

A local complementation on a vertex v of a graph G consists in complementing the subgraph induced by the neighborhood of v, more precisely, it leads to a graph $G \star v$ such that $V(G \star v) = V(G)$ and $E(G \star v) = E(G) \oplus E(K_{N_G(v)})$ where K_S denotes the complete graph on the vertices in S, and $\oplus$ denotes the symmetric difference of two sets. We say that G' is a vertex-minor of G, if G' can be obtained from G by means of local complementations and vertex deletions. Here we consider $V(G') \subseteq V(G)$ and require G' to be obtained exactly (not up to an isomorphism of graphs), meaning that there exists a sequence of graph transformations consisting of local complementations and the deletions of the vertices of $V(G) \setminus V(G')$.

Definition 2.1. Given a graph G, a vertex subset $V' \subseteq V(G)$, and an integer $k > 0$, we say that:

- G is k-vertex-minor universal on V', if $k \leqslant |V'|$ and any graph on any k vertices in V' is a vertex-minor of G.
- G is k-pairable on V', if $k \leqslant |V'|/2$ and any pairing on any $2k$ vertices in V' is a vertex-minor of G.

If any of the above properties is satisfied with $V' = V(G)$, we say that G is k-vertex-minor universal or that G is k-pairable, respectively.

Definition 2.2. We say that a bipartite graph $G = (L, R, E)$ is left (resp. right) k-vertex-minor universal or k-pairable if the corresponding condition from Definition 2.1 is satisfied for $V' = L$ (resp. $V' = R$). We say that G is two-side k-vertex-minor universal / k-pairable if it is both left and right k-vertex-minor universal / k-pairable.

Graph states form a standard family of quantum states that can be represented using simple undirected graphs (Ref. [16] is an excellent introduction to graph states). Given a graph $G = (V, E)$, the corresponding graph state $|G\rangle$ is the $|V|$-qubit state:

$$|G\rangle = \frac{1}{\sqrt{2}^{|V|}} \sum_{x \in 2^V} (-1)^{|G[x]|} |x\rangle$$

where $|G[x]|$ is the size (number of edges) of the subgraph induced by x [4].

We shall alternatively refer to the vertex set V as qubit set. A graph state $|G\rangle$ can be prepared as follows: initialize every qubit in $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ then apply for each edge of the graph a CZ gate on the corresponding pair of qubits, where $CZ : |ab\rangle \mapsto (-1)^{ab}|ab\rangle$. The graph state $|G\rangle$ is the unique quantum state (up to a global phase) that, for every vertex $u \in V$, is a fixed point of the Pauli operator $X_u Z_{N_G(u)}$ [5]. Hence, graph states form a subfamily of stabilizer states. Formally, an n-qubit stabilizer state [14] is a quantum state that is the simultaneous eigenvector with eigenvalue 1 of n commuting and independent Pauli operators. A useful property is that any stabilizer state is related to some graph state by the application of local Clifford unitaries, and these unitaries can be computed efficiently [10]. For instance, an EPR pair is equal to $|K_2\rangle$ up to local Clifford unitaries, where K_2 is the graph with two vertices and one edge. Thus, it is equivalent to be able to generate any stabilizer state, or any graph state, on a given set of qubits, by means of LOCC protocols. We introduce below the notion of k-stabilizer universal states.

Definition 2.3. A quantum state $|\psi\rangle$ is k-stabilizer universal (resp., k-pairable) if any stabilizer state on any k qubits in V (resp., any k EPR pairs on any $2k$ qubits in V) can be induced by means of LOCC protocols.

If H is a vertex-minor of a G then the graph state $|H\rangle$ can be obtained from $|G\rangle$ using only local Clifford operations, local Pauli measurements and classical communications, and the converse is true when H has no isolated vertices [9]. As a pairing on $2k$ vertices has no isolated vertices, we have the following:

[4] With a slight abuse of notation we identify a subset (say $x = \{u_2, u_4\}$) of the set of qubits $V = \{u_1, \ldots, u_5\}$ with its characteristic binary word ($x = 01010$).

[5] It consists in applying $X : |a\rangle \mapsto |1-a\rangle$ on u and $Z : |a\rangle \mapsto (-1)^a|a\rangle$ on each of its neighbors in G.

Proposition 2.1. A graph G is k-pairable if and only if the corresponding graph state $|G\rangle$ is k-pairable using only local Clifford operations, local Pauli measurements, and classical communication.

In the case of vertex-minor universality and stabilizer universality, the characterization from [9] does not apply directly, because of possible isolated vertices. For instance, K_2 is not 2-vertex-minor universal since no local complementation can turn it into an empty graph. However, $|K_2\rangle$ is 2-stabilizer universal: with e.g. an X-measurement on each qubit, one can map the corresponding graph state (a maximally entangled pair of qubits) to the graph state composed a tensor product of two qubits. To be able to state a characterization, a solution is to introduce destructive measurements.

Proposition 2.2. Given two graphs G and H such that $V(H) \subseteq V(G)$, H is a vertex-minor of G if and only if $|H\rangle$ can be obtained from $|G\rangle$ (on the qubits corresponding to $V(H)$) using only local Clifford operations, local destructive Pauli measurements, and classical communications.

Proof. Notice that a similar statement – involving non-destructive measurements and only valid when H does not contain isolated vertices – has been proved in [9] (Theorem 2.2). We provide here a direct proof of Proposition 2.2 which is actually slightly simpler thanks to the use of destructive measurements. In the following proof all measurements are destructive. ($\Rightarrow$) Local complementations can be implemented by means of local Clifford unitaries, and vertex deletions by means of Z-measurements together with classical communications and Pauli corrections [10]. ($\Leftarrow$) We prove the property by induction on the number of measurements. If there are no measurements the property is true [10]. Otherwise, let u be the first qubit to be measured. Assume u is measured according to P and C_u is the Clifford operator applied on u before the measurement. $C_u^\dagger P C_u$ is proportional to some Pauli operator $P_0 \in \{X, Y, Z\}$:
(i) if $P_0 = Z$, then the measurement of u can be interpreted as a vertex deletion and leads to $|G \setminus u\rangle$ up to Pauli corrections, so, by induction hypothesis, H is a vertex minor of $G \setminus u$, so is of G.
(ii) if $P_0 = Y$, then the measurement of u can be interpreted as a Z-measurement on $|G \star u\rangle$ (up to a Clifford operator on some other qubits), thus according to (i), H is a vertex minor of $G \star u$, so is of G.
(iii) if $P_0 = X$ and $N_G(u) \neq \emptyset$ then the measurement u can be interpreted as a Y-measurement on $|G \star v\rangle$ with $v \in N_G(u)$ (up to local Clifford operations on qubits different from u), thus according to (ii) H is a vertex minor of $G \star v$, so is of G.
(iv) if $P_0 = X$ and $N_G(u) = \emptyset$ then $|G\rangle = |G \setminus u\rangle \otimes |+\rangle_u$ so after the measurement of u the state is $|G \setminus u\rangle$, thus, by induction hypothesis, H is a vertex minor of $G \setminus u$, so is of G. □

Corollary 2.1. A graph G is k-vertex-minor universal if and only if the corresponding graph state $|G\rangle$ is k-stabilizer universal using only local Clifford operations, local destructive Pauli measurements, and classical communication.

Relations between pairability, vertex-minor universality and stabilizer universality of graph and graph states, can be found in Figure 1. To the best of our knowledge, all known examples of k-stabilizer universal (resp. k-pairable) graph

states come from k-vertex-minor universal (resp. k-pairable) graphs. Furthermore, to date, it is not known whether there exist k-pairable states which are not $2k$-stabilizer universal. Throughout this paper, we will essentially focus on the existence and the explicit construction of k-vertex-minor universal graphs.

$$\begin{array}{ccc} G \text{ is } 2k\text{-vertex-minor universal} & \Longrightarrow & |G\rangle \text{ is } 2k\text{-stabilizer universal} \\ \Downarrow & & \Downarrow \\ G \text{ is } k\text{-pairable} & \Longrightarrow & |G\rangle \text{ is } k\text{-pairable} \end{array}$$

Fig. 1: Implications between pairability, vertex-minor universality and stabilizer universality of graphs and graph states.

3 Existence of k-vertex-minor universal graphs of order quadratic in k

Given any k, a k-vertex-minor universal graph has at least a quadratic order in k:

Proposition 3.1 ([5]). If a graph G of order n is k-vertex-minor universal then $k<\sqrt{2n\log_2(3)}+2$.

In this section we prove that this bound is tight asymptotically, i.e. there exists k-vertex-minor universal graphs whose order grows quadratically with k. This greatly improves over the probabilistic construction of [5], where the existence of k-vertex-minor universal graphs of order $O(k^4\ln(k))$ was proven.

Theorem 3.1. For any constant $\alpha > 2$, there exists k_0 s.t. for any $k > k_0$, there exists a k-vertex-minor universal graph G of order at most αk^2.

The remaining of this section is a proof of Theorem 3.1. First we bound the probability that some graph of order k is not a vertex-minor of a random bipartite graph G, in Lemma 3.1. Then we bound the probability that such a random bipartite graph is k-vertex-minor universal, in Lemma 3.2, by defining some algorithm that tries to generate any graph as a vertex-minor of G. Finally, we prove that there exists a k-vertex-minor universal bipartite graph of quadratic order in k. More precisely, the probability of a random bipartite graph of quadratic order being k-vertex-minor universal goes to 1 exponentially fast in k:

Proposition 3.2. Fix constants $\epsilon > 0$, $c > 2$, and $c' > \frac{1+\epsilon}{\ln(2)}$. There exists k_0 s.t. for any $k > k_0$, the random bipartite graph G (the probability of an edge existing between two vertices, one in $L(G)$ and one in $R(G)$, is $1/2$, independently of the other edges) with $|L(G)| = \lfloor c'k\ln(k)\rfloor$ and $|R(G)| = \lfloor ck^2\rfloor$, is k-vertex-minor universal with probability at least $1 - e^{-\epsilon k\ln(k)}$.

Proposition 3.2 will be proved alongside Theorem 3.1 in this section. Notationwise, given a set A and an integer k, $\binom{A}{k}$ refers to $\{B \subseteq A \mid |B| = k\}$.

Lemma 3.1. Consider a random bipartite graph G with $|L(G)| \geqslant k$, $|R(G)| \geqslant 4\binom{k}{2} + 5$: the probability of an edge existing between two vertices (one in $L(G)$ and one in $R(G)$) is $1/2$, independently of the other edges. Take $k \in \mathbb{N}$ and consider a set of vertices $K \in \binom{L(G)}{k}$. The probability that there exists a graph defined on K which is not a vertex-minor of G is upper bounded by $e^{-\frac{\left(\frac{|R(G)|}{4} - \binom{k}{2}+1\right)^2}{\left(\frac{7|R(G)|}{4} - \binom{k}{2}+1\right)}}$.

Proof. For some $j \in \mathbb{N} \setminus \{0\}$ and $X \in \binom{R(G)}{j}$, consider the incidence matrix M_X of size $j \times \binom{k}{2}$, whose column number i represent the pairs of vertices of K that are in the neighborhood of the ith vertex of X, in the sense that its entries are 1 if the pair of vertices u,v is in its neighborhood, 0 else. Note that if there exists some $X \in \binom{R(G)}{\binom{k}{2}}$ whose incidence matrix M_X is of full column-rank, then any $2^{\binom{k}{2}}$ graph defined on K is a vertex-minor of G. Indeed, column number i represents the edges (resp. non-edges) of K to be toggled by a local complementation on the ith vertex of X. So now we will bound the probability of such a set X existing within $R(G)$.

For this purpose we will greedily try to construct the set $X \in \binom{R(G)}{\binom{k}{2}}$, one vertex after the other, by considering each vertex in $R(G)$ one by one, and we will lower bound the probability of the event "there exists some $X \in \binom{R(G)}{\binom{k}{2}}$ whose incidence matrix M_X if of full column-rank" by the probability of success of the algorithm. The algorithm works as follows. Arbitrarily order the vertices of $R(G)$. At each step (say that we have j vertices in X at some step), suppose the corresponding matrix of incidence (of size $j \times \binom{k}{2}$) full column-rank. We consider the next vertex $u \in R(G)$ in the list: if adding its corresponding vector to M_X increases its column-rank, then we add u to X, else we remove u from the vertices to consider. The algorithm stops (and succeeds) if M_X has $\binom{k}{2}$ columns and is full column-rank. Let us show that the probability of a vertex u increasing the column-rank of M_X (if $j < \binom{k}{2}$) is lower-bounded by $1/4$.

If M_X is of rank $j < \binom{k}{2}$, there exists a non-zero vector W (i.e. a set of pairs of vertices of K) which is orthogonal to all j first vectors. W can be seen as the characteristic function of the edges of some graph H on the vertices of $L(G)$. Adding a vertex u to X increases the rank of M_X if the vector U of incidence of u in K is such that $U \cdot W = 1 \mod 2$ (because then U is not in the span of M_X). Note that, if H has exactly one edge, then there is exactly probability $\frac{1}{4}$ that $U \cdot W = 1 \mod 2$ (in this case the two ends of the unique edge of H are connected to u, which happens with probability $\frac{1}{2} \times \frac{1}{2}$). As H has at least one edge, it has at least one vertex of non-zero degree z. Let us draw randomly the neighborhood of u: first we draw among the vertices of $H \setminus \{z\}$, then we add z with probability $\frac{1}{2}$. The probability that an odd number of neighbors of z are neighbors of u is $1/2$, so drawing z changes the parity of the number of edges in H whose ends are both neighbors of u, with probability $1/2$. At the end of the day there is a probability of at least $\frac{1}{4}$ that $U \cdot W = 1 \mod 2$, so that u increases the column-rank of M_X.

Finally, the algorithm fails if we encounter more than $|R(G)| - \binom{k}{2} + 1$ vertices that did not increase the column-rank of M_X. Let us introduce a random variable T that follows the distribution $B(|R(G)|, 3/4)$. The probability that the algorithm fails is upper bounded by $\Pr(T \geqslant |R(G)| - \binom{k}{2} + 1)$. We'll use the

Chernoff bound: With $\mu = \mathbb{E}[T] = \frac{3|R(G)|}{4}$, for any $\delta > 0$, $\Pr(T \geqslant (1+\delta)\mu) \leqslant e^{-\frac{\delta^2}{2+\delta}\mu}$. As we need $(1+\delta)\mu = |R(G)| - \binom{k}{2} + 1$, we take $\delta = \frac{|R(G)|-\binom{k}{2}+1-\mu}{\mu} > 0$ (thanks to the conditions on $|R(G)|$). The Chernoff bound then gives

$$\Pr\left(T \geqslant |R(G)| - \binom{k}{2} + 1\right) \leqslant e^{-\frac{\left(\frac{|R(G)|-\binom{k}{2}+1-\mu}{\mu}\right)^2}{\left(\frac{|R(G)|-\binom{k}{2}+1+\mu}{\mu}\right)}\mu} = e^{-\frac{\left(|R(G)|-\binom{k}{2}+1-\mu\right)^2}{\left(|R(G)|-\binom{k}{2}+1+\mu\right)}} = e^{-\frac{\left(\frac{|R(G)|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7|R(G)|}{4}-\binom{k}{2}+1\right)}}$$

So the probability of the existence of $X \subseteq \binom{R(G)}{k}$ whose incidence matrix M_X if of full column-rank is lower bounded by $1 - e^{-\frac{\left(\frac{|R(G)|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7|R(G)|}{4}-\binom{k}{2}+1\right)}}$.

□

Lemma 3.2. Consider a random bipartite graph G with $|L(G)| \geqslant k$, $|R(G)| \geqslant 4\binom{k}{2} + 5$: the probability of an edge existing between two vertices (one in $L(G)$ and one in $R(G)$) is $1/2$, independently of the other edges. The probability that G is k-vertex-minor universal is lower bounded by

$$1 - \left(\frac{k}{2^{|L(G)|-k+1}} + e^{-\frac{\left(\frac{|R(G)|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7(|R(G)|-k)}{4}-\binom{k}{2}+1\right)}}\right) \times \binom{|L(G)|+|R(G)|}{k}$$

Proof. Given a set $K \in \binom{V}{k}$, we consider the bad event A_K: "there exists a graph defined on K which is not a vertex-minor of G". The probability that G is k-vertex-minor universal is, by definition, $\Pr\left(\bigcap_{K\in\binom{V}{k}} \overline{A_K}\right)$. Suppose that each probability $\Pr(A_K)$ is upper bounded by some p. Then, using the union bound,

$$\Pr\left(\bigcap_{K\in\binom{V}{k}} \overline{A_K}\right) = 1 - \Pr\left(\bigcup_{K\in\binom{V}{k}} A_K\right) \geqslant 1 - \sum_{K\in\binom{V}{k}} \Pr(A_K) \geqslant 1 - p \times \binom{|L(G)|+|R(G)|}{k}$$

Now, fix some $K \in \binom{V}{k}$. Let us upper bound $\Pr(A_K)$. For this purpose, let us show in the following how one can induce with high probability any graph on K as a vertex-minor, by first turning the graph G into a graph G' defined on a subset of V, which is bipartite (as G), and such that the vertices of K are all "on the left". Let $L_{\overline{K}} = L(G) \setminus (K \cap L(G))$, $R_K = R(G) \cap K$. The algorithm works as follows. Roughly speaking, we use pivotings to move the vertices of R_K from the right side to the left side. Given $a \in R(G)$ and $b \in L(G)$, pivoting an edge ab in a bipartite graph G produces the bipartite graph $G \wedge ab = G * a * b * a$ where the edges between $N_G(a) \setminus \{b\}$ and $N_G(b) \setminus \{a\}$ are toggled and vertices a and b are exchanged (in other words $N_{G\wedge ab}(b) = N_G(a)\Delta\{a,b\}$, $N_{G\wedge ab}(a) = N_G(b)\Delta\{a,b\}$, so that the graph is bipartite according to the partition $R' := R(G)\Delta\{a,b\}$, $L' := L(G)\Delta\{a,b\}$). Once all vertices of R_K are moved to the left by means of pivotings, we then consider the induced subgraph which consists in removing the vertices that have been moved from the left to the right side. We obtain a bipartite graph such that all vertices of K are on the left side, the idea

is then to apply Lemma 3.1 to show that with high probability one can induce any graph on K as a vertex-minor (using only local complementation on vertices on the right side). So we have to prove that the constructed graph behaves as a random bipartite graph: each edge exists independently with probability $1/2$.

To this end we provide a little more details on the algorithm. Given the initial random bipartite graph G, we proceed as follows: given a vertex $a \in R_K$, if there is no edge between a and $L_{\overline{K}}$, the algorithm fails. Otherwise, we consider an arbitrary vertex $b \in N_G(a) \cap L_{\overline{K}}$ and perform a pivoting on ab, then remove vertex b, leading to a graph $G \wedge ab \setminus b$ which is bipartite according to $R' := R \setminus \{a\}$, $L' := L(G)\Delta\{a, b\}$. We show in the following that this bipartite graph is random, i.e. each edge exists independently with probability $1/2$.

- For any $u \in R'$ we have $a \sim_{G\wedge ab} u$ if and only if $b \sim_G u$, so $\Pr(a \sim_{G\wedge ab} u) = \frac{1}{2}$.
- For any $u \in R', v \in L' \setminus \{a\}$ we have $u \sim_{G\wedge ab} v$ if and only if ($u \sim_G v$ XOR $(u \in N_G(b) \wedge v \in N_G(a))$). As the event $u \in N_G(b) \wedge v \in N_G(a)$ is independent of the existence of an edge between u and v in G, we have $\Pr(u \sim_{G\wedge ab} v) = \frac{1}{2}$.

Regarding independence, notice that the existence of an edge (u, v) in $G \wedge ab \setminus b$ is independent of the existence of all edges but (u, v) in G. The independence of the existence of the edges in G guarantees the independence in $G \wedge ab \setminus b$.

To sum up, starting from a random bipartite graph and a vertex $a \in R_K$, if there is an edge between a and some vertex $b \in L_{\overline{K}}$, we move a to the left side (by means of a pivoting) and remove b: the remaining graph is a random bipartite graph. The algorithm consists in repeating this process until R_K is empty, leading to a random bipartite graph. If the procedure succeeds, we end up with a bipartite graph G' with $|R(G')| = |R(G)| - |R_K|$, $|L(G')| = |L(G)|$ such that $R(G') \subseteq R(G)$ and $L(G') \subseteq L(G) \cup R_K$. Recall that we obtained G' from G using only local complementations and vertex-deletions, so any vertex-minor of G' is a vertex-minor of G. At the end of the day, $\Pr(A_K)$ is upper bounded by the probability that (1) the algorithm fails or (2) there exists a graph defined on K which is not a vertex-minor of G'.

- (1): Let us upper bound the probability that the algorithm fails. The algorithm is composed of $|R_K|$ steps. At each step, the algorithm fails if there is no edge between a and $L_{\overline{K}}$. The bipartite graph, at this point, is random: each edges between a and the vertices of $L_{\overline{K}}$ exist independently with probability $1/2$. The probability that there is no edge between a and $L_{\overline{K}}$ is $\frac{1}{2^{|L_{\overline{K}}|}}$. In general, $|L_{\overline{K}}|$ is lower bounded by $|L(G)|-k+1$, and $|R_K|$ is upper bounded by k. At the end of the day, using the union bound, the algorithm fails with probability at most $\frac{k}{2^{|L(G)|-k+1}}$.
- (2): Suppose the algorithm succeeds. We end up with a random bipartite graph G'. Using Lemma 3.1, the probability that there exists a graph defined on K which is not a vertex-minor of G' is upper bounded by

$$e^{-\frac{\left(\frac{|R(G)|-|R_K|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7(|R(G)|-|R_K|)}{4}-\binom{k}{2}+1\right)}} \leqslant e^{-\frac{\left(\frac{|R(G)|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7(|R(G)|-k)}{4}-\binom{k}{2}+1\right)}}$$

So, at the end of the day,

$$\Pr(A_K) \leqslant \frac{k}{2^{|L(G)|-k+1}} + e^{-\frac{\left(\frac{|R(G)|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7(|R(G)|-k)}{4}-\binom{k}{2}+1\right)}}$$

And then,

$$\Pr\left(\bigcap_{K\in\binom{V}{k}} \overline{A_K}\right) \leqslant 1-\left(\frac{k}{2^{|L(G)|-k+1}} + e^{-\frac{\left(\frac{|R(G)|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7(|R(G)|-k)}{4}-\binom{k}{2}+1\right)}}\right)\times\binom{|L(G)|+|R(G)|}{k}$$

□

Remark 3.1. Lemma 3.2 has concrete applications on its own right: in particular for any integer k, it yields an integer n such that there exists a (bipartite) k-vertex-minor universal graph of order n. In general, one can infer a lower bound on the probability of generating a k-vertex-minor universal graph, for any choice of k and n, using the algorithm presented in the proof of Lemma 3.2. A table presenting orders for which some bipartite k-vertex-minor universal graph exists, as well as orders for with a randomly generated bipartite graph is k-vertex-minor universal with at least 99% probability, for particular values of k ranging from 3 to 100, can be found in Section A. Surprisingly enough, we observe that a small constant additive overheard in the order of the graph is sufficient to attain a high probability of generating a k-vertex-minor universal graph.

Now we are ready to conclude. Fix some constants $c > 2$ and $c' > \frac{1}{\ln(2)}$. Let G be a random bipartite graph G with $|L(G)| = \lfloor c'k\ln(k)\rfloor$ and $|R(G)| = \lfloor ck^2\rfloor$: the probability of an edge existing between two vertices (one in $L(G)$ and one in $R(G)$) is $1/2$, independently of the other edges.

Note $n = |V| = |L(G)| + |R(G)| = \lfloor c'k\ln(k)\rfloor + \lfloor ck^2\rfloor$. Using Lemma 3.2, the probability that G is k-vertex-minor universal is lower bounded by

$$1 - \left(\frac{k}{2^{|L(G)|-k+1}} + e^{-\frac{\left(\frac{|R(G)|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7(|R(G)|-k)}{4}-\binom{k}{2}+1\right)}}\right)\times\binom{n}{k}$$

Let us prove that this probability is positive with our choice of parameters, for some big enough k. It is sufficient to have:

$$(1)\quad \frac{k}{2^{|L(G)|-k+1}}\binom{n}{k} < \frac{1}{2} \quad\text{and}\quad (2)\quad e^{-\frac{\left(\frac{|R(G)|}{4}-\binom{k}{2}+1\right)^2}{\left(\frac{7(|R(G)|-k)}{4}-\binom{k}{2}+1\right)}}\binom{n}{k} < \frac{1}{2}$$

Let us show that these equations are satisfied for any large enough k. Recall that $\binom{n}{k} \leqslant 2^{nH(k/n)}$ where $H(x) = -x\log_2(x) - (1-x)\log_2(1-x)$ is the binary entropy.

(1): It is sufficient that $\log_2(k) + nH(k/n) - |L(G)| + k - 1 < -1$.

$\log_2(k)+nH(k/n)-|L(G)|+k-1 \sim_{k\to\infty} n\frac{k}{n}\log_2(\frac{k}{n})-c'k\ln(k) = k(\log_2(k)-\log_2(n)) - c'k\ln(k) \sim_{k\to\infty} \frac{1}{\ln(2)}k\ln(k) - c'k\ln(k)$. The choice of c' guarantees that for any large enough k, (1) is satisfied.

(2): It is sufficient that $nH(k/n)\ln(2) - \frac{\left(\frac{|R(G)|}{4} - \binom{k}{2} + 1\right)^2}{\left(\frac{7(|R(G)|-k)}{4} - \binom{k}{2} + 1\right)} < -\ln(2) \cdot \frac{\left(\frac{|R(G)|}{4} - \binom{k}{2} + 1\right)^2}{\left(\frac{7(|R(G)|-k)}{4} - \binom{k}{2} + 1\right)}$ $\sim_{k\to\infty} \frac{\left(\frac{ck^2}{4} - \frac{k^2}{2}\right)^2}{\left(\frac{7ck^2}{4} - \frac{k^2}{2}\right)} = k^2 \frac{(c-2)^2}{4(7c-2)}$. We saw above that $nH(k/n)\ln(2) \sim_{k\to\infty} k\ln(k)$. The choice of c guarantees that for any large enough k, (1) is satisfied.

This proves that, for any large enough k, G of order $\lfloor c'k\ln(k)\rfloor + \lfloor ck^2\rfloor$, is k-vertex-minor universal with non-zero probability. Taking $\alpha > c$, for any large enough k, $\lfloor c'k\ln(k)\rfloor + \lfloor ck^2\rfloor \leqslant \alpha k^2$, proving Theorem 3.1.

Furthermore, we just saw that side (1) of the equation dominates (2) asymptotically. Thus, the probability of G being k-vertex-minor universal is roughly lower bounded by $1 - 2^{\frac{1}{\ln(2)}k\ln(k) - c'k\ln(k)} = 1 - e^{-(ln(2)c'-1)k\ln(k)}$ as k grows. Then, for any $\epsilon > 0$ such that $\epsilon < \ln(2)c' - 1$, for any large enough k, G of order $\lfloor c'k\ln(k)\rfloor + \lfloor ck^2\rfloor$, is k-vertex-minor universal with probability at least $1 - e^{-\epsilon k\ln(k)}$, proving Proposition 3.2.

4 Vertex-minor universal graphs from projective planes

In this section, we provide explicit constructions of families of k-vertex-minor universal graphs, of order n proportional to k^4. Thus, the order of the constructed graphs scales as the square of the asymptotically optimal graph order from Section 3. To the best of our knowledge, these are the first explicit constructions of k-vertex-minor universal (or k-pairable) graphs of order n polynomial in k.

We start in Section 4.1 with some preparatory lemmas. In Section 4.2, we introduce a family of bipartite incidence graphs of projective planes, and study their k-pairability and k-vertex-minor universality properties. In Section 4.3 we introduce a new family of so-called reduced graphs from projective planes, and investigate their k-vertex-minor universality properties. A brief comparison between the two constructions is provided in Section 4.4.

4.1 Sufficient conditions for k-pairability and k-vertex-minor universality

Below and throughout Section 4, given a graph G, a vertex $v \in V(G)$, and a vertex subset $U \subseteq V(G)$, we shall use the shorthand notation $N_U(v) := N_G(v) \cap U$, that is, the set of neighbors of v that belong to U (in such a case, we shall always ensure that the context makes the choice of G unambiguous).

The following two lemmas give sufficient conditions for a bipartite graph G to be one-side (i.e., left or right) k-pairable or k-vertex-minor universal. For simplicity, we state these conditions for the set of left vertices.

Lemma 4.1. Let G be a bipartite graph satisfying the following property:

(P) For any set of $2k$ vertices $K = \{u_1, v_1, u_2, v_2, \ldots, u_k, v_k\} \subseteq L(G)$, there exist:

  (i) a set of k vertices $C = \{c_1, c_2, \ldots, c_k\} \subseteq L(G)$, with $C \cap K = \emptyset$, and

  (ii) a set of $2k$ vertices $S = \{\alpha_1, \beta_1, \alpha_2, \beta_2, \ldots, \alpha_k, \beta_k\} \subseteq R(G)$, such that $N_{K\cup C}(\alpha_i) = \{u_i, c_i\}$ and $N_{K\cup C}(\beta_i) = \{v_i, c_i\}$, for all $i = 1, \ldots, k$.

Then G is left k-pairable.

Proof. We use first local complementation on vertices α_i and β_i to create edges (u_i, c_i) and (v_i, c_i), followed by local complementation on vertices c_i to create edges (u_i, v_i), as desired. It is easily seen that no edges are created between u_i and $K \setminus \{v_i\}$, or between v_i and $K \setminus \{u_i\}$. □

Lemma 4.2. Let G be a bipartite graph satisfying the following property:

(VMU) For any set of k vertices $K = \{u_1, u_2, \ldots, u_k\} \subseteq L(G)$, there exist:

(i) a set of $k(k-1)/2$ vertices $C = \{c_{ij} \mid 1 \leqslant i < j \leqslant k\} \subseteq L(G)$, with $C \cap K = \emptyset$, and

(ii) a set of $k(k-1)$ vertices $S = \{\alpha_{ij}, \beta_{ij} \mid 1 \leqslant i < j \leqslant k\} \subseteq R(G)$, such that $N_{K \cup C}(\alpha_{ij}) = \{u_i, c_{ij}\}$ and $N_{K \cup C}(\beta_{ij}) = \{u_j, c_{ij}\}$, for all $1 \leqslant i < j \leqslant k$.

Then G is left k-vertex-minor universal.

Proof. (See also Figure 2a.) The proof is similar to that of Lemma 4.1. To create an edge between u_i and u_j, we use first local complementation on vertices α_{ij} and β_{ij}, followed by local complementation on vertex c_{ij}. This procedure does not create any other edge between the vertices of K. □

Providing sufficient conditions for a bipartite graph G to be k-vertex-minor universal (on the entire vertex set) is more involved. To induce an arbitrary graph with vertex set $K = K_1 \sqcup K_2$, where $K_1 \subseteq L(G)$ and $K_2 \subseteq R(G)$, we may need to create edges with both endpoints in either K_1 or K_2, which can be dealt with by using conditions similar to those in Lemma 4.2, but also "toggle" (i.e., either create or remove, as needed) edges between K_1 and K_2, which represents an additional difficulty. We give sufficient conditions for doing so, in the lemma below (see also Figure 2b).

Lemma 4.3. Let G be a bipartite graph satisfying the following property:

(VMU*) For any set of k vertices $K = K_1 \sqcup K_2$, with $K_1 = \{u_1, \ldots, u_{k_1}\} \subseteq L(G)$, and $K_2 = \{\lambda_1, \ldots, \lambda_{k_2}\} \subseteq R(G)$, there exist:

(i) a subset $C_1 = \{c_{ij} \mid 1 \leqslant i < j \leqslant k_1\} \subseteq L(G)$, such that $C_1 \cap K_1 = \emptyset$ and $N_{K_2}(C_1) = \emptyset$,

(ii) a subset $S_1 = \{\alpha_{ij}, \beta_{ij} \mid 1 \leqslant i < j \leqslant k_1\} \subseteq R(G)$, such that $S_1 \cap K_2 = \emptyset$ and
for all $1 \leqslant i < j \leqslant k_1, N_{K_1 \sqcup C_1}(\alpha_{ij}) = \{u_i, c_{ij}\}$ and $N_{K_1 \sqcup C_1}(\beta_{ij}) = \{u_j, c_{ij}\}$,

(iii) a subset $\Omega = \{\omega_{ij} \mid 1 \leqslant i \leqslant k_1, 1 \leqslant j \leqslant k_2\} \subseteq R(G)$ such that $\Omega \cap (K_2 \sqcup S_1) = \emptyset$ and
for all $1 \leqslant i \leqslant k_1, 1 \leqslant j \leqslant k_2, N_{K_1 \sqcup C_1}(\omega_{ij}) = \{u_i\}$,

(iv) a subset $C_2 = \{\gamma_{ij} \mid 1 \leqslant j \leqslant k_2, j < i \leqslant k_1 + k_2\} \subseteq R(G)$ such that $C_2 \cap (K_2 \sqcup S_1 \sqcup \Omega) = \emptyset$ and $N_{K_1 \sqcup C_1}(C_2) = \emptyset$,

(v) a subset $S_2 = \{a_{ij}, b_{ij} \mid 1 \leqslant j \leqslant k_2, j < i \leqslant k_1 + k_2\} \subseteq L(G)$ such that $S_2 \cap (K_1 \sqcup C_1) = \emptyset$ and for all $1 \leqslant j \leqslant k_2$, $j < i \leqslant k_1 + k_2$,
• $N_{K_2 \sqcup S_1 \sqcup \Omega \sqcup C_2}(a_{ij}) = \{\lambda_i, \gamma_{ij}\}$ and $N_{K_2 \sqcup S_1 \sqcup \Omega \sqcup C_2}(b_{ij}) = \{\lambda_j, \gamma_{ij}\}$, if $i \leqslant k_2$
• $N_{K_2 \sqcup S_1 \sqcup \Omega \sqcup C_2}(a_{ij}) = \{\omega_{(i-k_2)j}, \gamma_{ij}\}$ and $N_{K_2 \sqcup S_1 \sqcup \Omega \sqcup C_2}(b_{ij}) = \{\lambda_j, \gamma_{ij}\}$, otherwise.

Then G is k-vertex-minor universal.

Proof. We start by removing all vertices that are not in any set defined in (VMU*). Then we proceed in the following three steps.

1) In case we need to create an edge (u_i, u_j) for $1 \leqslant i < j \leqslant k_1$ between two vertices in K_1. We first use local complementations on α_{ij} and β_{ij} to create edges (u_i, c_{ij}) and (u_j, c_{ij}) (no other edges are created) and then remove α_{ij} and β_{ij}. Then, we use local complementation on c_{ij} to create the edge (u_i, u_j) (no other edges are created). Finally, we remove vertex c_{ij}, thus only the edge (u_i, u_j) has been constructed.

2) In case we need to create an edge (λ_i, λ_j) for $1 \leqslant j < i \leqslant k_2$ between two vertices in K_2. We first use local complementations on a_{ij} and b_{ij} to create edges (λ_i, γ_{ij}) and (λ_j, γ_{ij}) (no other edges are created) and then remove a_{ij} and b_{ij}. Then, we use local complementation on γ_{ij} to create the edge (λ_i, λ_j) (no other edges are created). Finally, we remove vertex γ_{ij}, thus only the edge (λ_i, λ_j) has been constructed.

3) In case we need to toggle an edge (u_i, λ_j) for $1 \leqslant i \leqslant k_1$ and $1 \leqslant j \leqslant k_2$ between two vertices in K_1 and K_2. We first use local complementations on $a_{(i+k_2)j}$ and $b_{(i+k_2)j}$ to create edges $(\omega_{ij}, \gamma_{(i+k_2)j})$ and $(\lambda_j, \gamma_{(i+k_2)j})$ (no other edges are created) and then remove $a_{(i+k_2)j}$ and $b_{(i+k_2)j}$. Then, we use local complementation on $\gamma_{(i+k_2)j}$ to create the edge (ω_{ij}, λ_j) (no other edges are created). After that, we remove vertex γ_{ij}, thus only the edge (ω_{ij}, λ_j) has been constructed. Finally, we use local complementation on ω_{ij} to create the edge (u_i, λ_j) (no other edges are created). Then, we remove vertex ω_{ij}, thus only the edge (u_i, λ_j) has been toggled. □

The following lemma is a generalization of Lemma 4.2 to the case of general (not necessarily bipartite) graphs.

Lemma 4.4. Let G be a graph satisfying the following property:

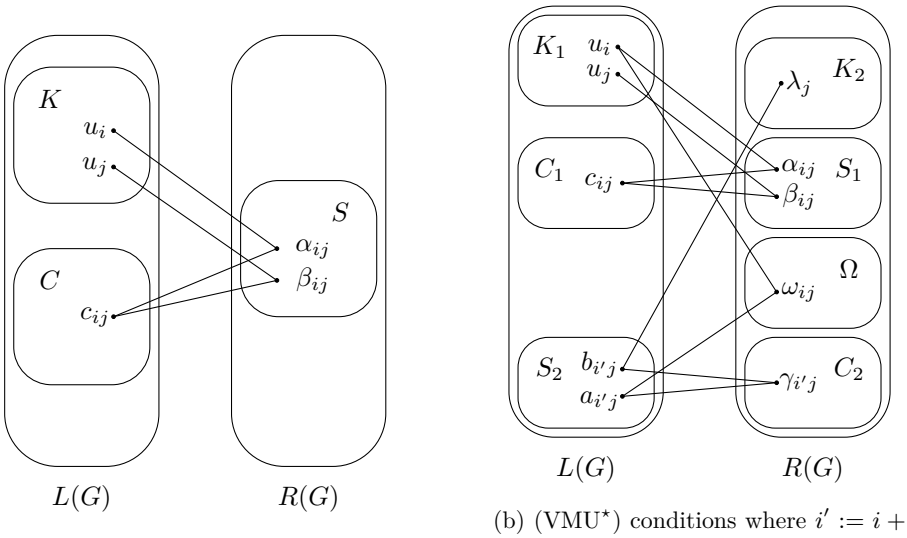
(VMU$^\circ$) For any set of k vertices $K = \{u_1, u_2, \ldots, u_k\} \subseteq V(G)$, there exist:

(i) a set of $k(k-1)/2$ vertices $C = \{c_{ij} \mid 1 \leqslant i < j \leqslant k\} \subseteq V(G)$, such that C is stable, $C \cap K = \emptyset$, and $N_K(c_{ij}) = \emptyset$, for all $1 \leqslant i < j \leqslant k$, and

(ii) a set of $k(k-1)$ vertices $S = \{a_{ij}, b_{ij} \mid 1 \leqslant i < j \leqslant k\} \subseteq V(G)$, such that S is stable, $S \cap (K \cup C) = \emptyset$, $N_{K \cup C}(a_{ij}) = \{u_i, c_{ij}\}$ and $N_{K \cup C}(b_{ij}) = \{u_j, c_{ij}\}$, for all $1 \leqslant i < j \leqslant k$.

Then G is k-vertex-minor universal.

Proof. Whenever we need to create or to remove an edge between vertices $u_i, u_j \in K$, we use first local complementation on vertices a_{ij} and b_{ij} to create edges between u_i and c_{ij}, and between u_j and c_{ij}, and then we use local complementation on c_{ij}. □



(a) (VMU) conditions.

(b) (VMU*) conditions where $i' := i + k_2$.

Fig. 2: Illustration of the (VMU) and (VMU*) conditions from Lemma 4.2 and Lemma 4.3.

4.2 Bipartite graphs from projective planes

Let $q > 0$ be a prime power, $\mathbb{F}_q$ be the finite field with q elements, and $\mathrm{PG}(2,q) := \left(\mathbb{F}_q^3\right)^* / \mathbb{F}_q^*$ be the projective plane over $\mathbb{F}_q$. Points and lines of $\mathrm{PG}(2,q)$ are identified respectively to 1-dimensional and 2-dimensional linear subspaces of $\mathbb{F}_q^3$. A line λ passes through a point a (we write $a \in \lambda$) if the 2-dimensional linear subspace of $\mathbb{F}_q^3$ corresponding to λ contains the 1-dimensional linear subspace corresponding to a. We will use the following properties of the projective plane:

- $\mathrm{PG}(2,q)$ has q^2+q+1 points and q^2+q+1 lines.
- Any line contains exactly $q+1$ points, and any point is contained in exactly $q+1$ lines.
- Any two distinct lines intersect in one point, and for any two distinct points there is one unique line containing them.

We denote by $\mathbb{G}_q$ the bipartite incidence graph of the projective plane $\mathrm{PG}(2,q)$. Precisely, the set of left vertices $L(\mathbb{G}_q)$ is the set of points of $\mathrm{PG}(2,q)$, the set of right vertices $R(\mathbb{G}_q)$ is the set of lines of $\mathrm{PG}(2,q)$, and the set of edges $E(\mathbb{G}_q)$ corresponds to incidences between points and lines, that is $E(\mathbb{G}_q) = \{(a,\lambda) \in L(\mathbb{G}_q) \times R(\mathbb{G}_q) \mid a \in \lambda\}$.

Theorem 4.1. Let k be such that $k \leqslant (q+4)/5$. Then $\mathbb{G}_q$ is two-side k-pairable.

Proof. Due to the symmetry of $\mathbb{G}_q$, it is enough to prove it is left k-pairable. For this, we will use Lemma 4.1. Let $K = \{u_1, v_1, u_2, v_2, \ldots, u_k, v_k\} \subseteq L(\mathbb{G}_q)$ be a set of $2k$ points. To construct the sets $C = \{c_1, c_2, \ldots, c_k\} \subseteq L(\mathbb{G}_q)$ and $S = \{\alpha_1, \beta_1, \alpha_2, \beta_2, \ldots, \alpha_k, \beta_k\} \subseteq R(\mathbb{G}_q)$ from the property (P) in Lemma 4.1, we will proceed by recursion.

First, since there are $q+1$ lines passing through u_1 and $|K \setminus \{u_1\}| = 2k-1 \leqslant q$, we may choose a line α_1 passing through u_1 and not passing through any other point in $K \setminus \{u_1\}$. Similarly, let β_1 be a line passing through v_1 and not passing through any other point in $K \setminus \{v_1\}$. We take c_1 to be the intersection point between α_1 and β_1.

For $1 \leqslant j < k$, assume that we have constructed a set of j points $C_j = \{c_1, \ldots, c_j\} \subseteq L(\mathbb{G}_q)$ and a set of $2j$ lines $S_j = \{\alpha_1, \beta_1, \ldots, \alpha_j, \beta_j\} \subseteq R(\mathbb{G}_q)$, satisfying the following conditions:

(i) $C_j \cap K = \emptyset$,

(ii) $N_{K \cup C_j}(\alpha_i) = \{u_i, c_i\}$ and $N_{K \cup C_j}(\beta_i) = \{v_i, c_i\}$, for all $i = 1, \ldots, j$.

To construct $\alpha_{j+1}, \beta_{j+1}$, and c_{j+1}, we proceed in the following steps (see also Figure 3).

- We take α_{j+1} to be any line passing through u_{j+1} and not passing through any other point in $(K \setminus \{u_{j+1}\}) \cup C_j$.
  This is possible since $|(K \setminus \{u_{j+1}\}) \cup C_j| = 2k-1+j \leqslant 3k-2 \leqslant q$. Moreover, $\alpha_{j+1} \notin S_j$, since by construction no line in S_j passes through u_{j+1}. We further denote by $I_{j+1} \subseteq L(\mathbb{G}_q)$ the set consisting of the intersection points between α_{j+1} and the $2j$ lines in S_j. Thus, $|I_{j+1}| \leqslant 2j$.

- We take β_{j+1} to be any line passing through v_{j+1} and not passing through any other point in $(K \setminus \{v_{j+1}\}) \cup C_j \cup I_{j+1}$.
  This is possible since $|(K \setminus \{v_{j+1}\}) \cup C_j \cup I_{j+1}| \leqslant 2k-1+3j \leqslant 5k-4 \leqslant q$. Clearly, $\beta_{j+1} \notin S_j \cup \{\alpha_{j+1}\}$, since no line in $S_j \cup \{\alpha_{j+1}\}$ passes through v_{j+1}.

- We take c_{j+1} to be the intersection point between α_{j+1} and β_{j+1}.
  Clearly, $c_{j+1} \notin C_j$, since neither one of α_{j+1} nor β_{j+1} passes through the points in C_j.

To complete our recursion, we need to prove:

(i) $C_{j+1} \cap K = \emptyset$. We only have to prove that $c_{j+1} \notin K$. This follows from the fact that each of α_{j+1} and β_{j+1} passes through only one point in K, namely u_{j+1} and v_{j+1}, respectively, and they are distinct.

(ii) $N_{K \cup C_{j+1}}(\alpha_i) = \{u_i, c_i\}$ and $N_{K \cup C_{j+1}}(\beta_i) = \{v_i, c_i\}$, for all $i = 1, \ldots, j+1$.
  For $i = j+1$, the above equalities follow by construction. Indeed, α_{j+1} passes through u_{j+1} and c_{j+1}, but it does not pass through any other point in $(K \setminus \{u_{j+1}\}) \cup C_j$, and similarly, β_{j+1} passes through v_{j+1} and c_{j+1}, but it does not pass through any other point in $(K \setminus \{v_{j+1}\}) \cup C_j$.
  For $1 \leqslant i \leqslant j$, we only need to prove that neither α_i nor β_i passes through c_{j+1}. This follows from the fact that β_{j+1} does not pass through any point of I_{j+1}. Indeed, assuming that c_{j+1} belongs to either α_i or β_i, implies it belongs to I_{j+1}, the set of intersection points between α_{j+1} and the lines in S_j. This contradicts the fact that β_{j+1} does not pass through any point of I_{j+1}.

By recursion, we can construct sets $C := C_k$ and $S := S_k$ satisfying the property (P) from Lemma 4.1, and thus we conclude that $\mathbb{G}_q$ is left k-pairable. □

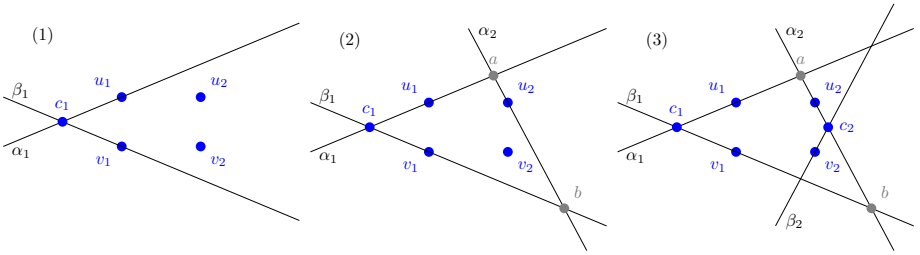


Fig. 3: Recursive construction of sets C and S in the proof of of Theorem 4.1, for $k = 2$. (1) We chose α_1 any line passing through u_1, and not passing through v_1, u_2, v_2. Similarly, we choose β_1 passing through v_1, and not passing through u_1, u_2, v_2. We take c_1 the intersection point between α_1 and β_1. (2) We chose α_2 any line passing through u_2, and not passing through u_1, v_1, c_1, v_2. We determine the intersection points a and b of α_2 with α_1 and β_1. (3) We chose β_2 any line passing through v_2, and not passing through u_1, v_1, c_1, u_2, as well as a, b (to avoid α_2 and β_2 intersecting on these points). We take c_2 the intersection point between α_2 and β_2.

Theorem 4.2. Let k be such that $3k^2 - k - 8 \leqslant 2q$. Then $\mathbb{G}_q$ is two-side k-vertex-minor universal.

Proof. Due to the symmetry of $\mathbb{G}_q$, it is enough to prove it is left k-vertex-minor universal. We prove $\mathbb{G}_q$ satisfies the property (VMU) from Lemma 4.2. Let $K = \{u_1, u_2, \ldots, u_k\} \subseteq L(\mathbb{G}_q)$ be a set of k points. To construct the sets $C = \{c_{ij} \mid 1 \leqslant i < j \leqslant k\} \subseteq L(\mathbb{G}_q)$ and $S = \{\alpha_{ij}, \beta_{ij} \mid 1 \leqslant i < j \leqslant k\} \subseteq R(\mathbb{G}_q)$ from Lemma 4.2 we will proceed again by recursion, by running through pairs (u_i, u_j) in some particular order, say in lexicographical order with respect to indexes (i, j).

The recursion is similar to the one in the proof of Lemma 4.1. We construct recursively lines α_{ij} and β_{ij}, passing through u_i and u_j, respectively, and take $c_{ij} = \alpha_{ij} \cap \beta_{ij}$. In the recursion, we take α_{ij} to be any line passing through u_i and not passing through any other point in $(K \setminus \{u_i\}) \cup C_{ij}$, where $C_{ij} := \{c_{i'j'} \mid (i', j') < (i, j)\}$. Since $|(K \setminus \{u_i\}) \cup C_{ij}| \leqslant (k-1) + (k(k-1)/2 - 1) = \frac{1}{2}(k^2 + k - 4)$, such a choice of α_{ij} is possible if $k^2 + k - 4 \leqslant 2q$.

A stronger constraint on the value of k comes from the choice of β_{ij}. Indeed, for β_{ij} we take any line passing through u_j and not passing through any other point in $(K \setminus \{u_j\}) \cup C_{ij} \cup I_{ij}$, where I_{ij} is the set of intersection points between α_{ij} and the previously constructed lines $\alpha_{i'j'}$ and $\beta_{i'j'}$, with $(i', j') < (i, j)$. Since $|(K \setminus \{u_j\}) \cup C_{ij} \cup I_{ij}| \leqslant (k-1) + 3(k(k-1)/2 - 1) = \frac{1}{2}(3k^2 - k - 8)$, we conclude that such a choice of β_{ij} is possible as long as $\frac{1}{2}(3k^2 - k - 8) \leqslant q$, as stated in the lemma. □

Theorem 4.3. Let k be such that $7k^2 - 16 \leqslant 4q$. Then $\mathbb{G}_q$ is k-vertex-minor universal.

Sketch of proof. Here, we will prove that $\mathbb{G}_q$ satisfies the property (VMU*) from Lemma 4.3. A first step is to build C_1 and S_1, used to construct edges, as needed, between vertices in K_1. This is done similarly as in the proof of Theorem 4.2,

except that they have to be disjoint from and not neighbors of K_2. Then, we choose Ω disjoint from $K_2 \sqcup S_1$, and such that the neighborhood of ω_{ij} intersects $C_1 \sqcup K_1$ in u_i only. Finally, the last step consists in building the sets S_2 and C_2, used to construct edges, as needed, between vertices in $K_2 \sqcup \Omega$. This is done again similarly as in the proof of Theorem 4.2 (with the role of left and right vertices inverted), and by taking into account that S_2 and C_2 have to be disjoint from and not neighbors of some of the previously constructed sets. The detailed proof is given in Section B. □

4.3 Reduced graphs from projective planes

Definition 4.1. Let G be a bipartite graph and $\varphi : L(G) \to R(G)$. The φ-reduction of G is the graph G_φ such that:

- The vertex set of G_φ is the left vertex set of G, that is $V(G_\varphi) = L(G)$,
- There is an edge between $a, b \in V(G_\varphi)$, if $a \neq b$ and either (a and $\varphi(b)$) or (b and $\varphi(a)$) are neighbors in G, that is,

$$E(G_\varphi) = \{(a,b) \mid a \neq b \text{ and } [\,(a,\varphi(b)) \in E(G) \text{ or } (b,\varphi(a)) \in E(G)\,]\,\}$$

The reduction is said to be bijective if φ is bijective. It is said to be symmetric if φ is such that $(a,\varphi(b)) \in E(G) \Leftrightarrow (b,\varphi(a)) \in E(G), \forall a, b \in L(G)$.

We enforce the condition $a \neq b$ in the definition of $E(G_\varphi)$, in order to avoid loops in case $(a,\varphi(a)) \in E(G)$ for some $a \in L(G)$.

Let G_φ be a bijective, symmetric reduction of G. For any vertex $a \in V(G_\varphi) = L(G)$, let $N_{G_\varphi}(a) \subseteq L(G)$ be the set of neighbors of a in G_φ, and $N_G(a) \subseteq R(G)$ be the set of neighbors of a in G. By definition, if $b \in N_{G_\varphi}(a)$ then $\varphi(b) \in N_G(a)$. The converse is also true, except if $\varphi(a) \in N_G(a)$, or equivalently, $(a,\varphi(a)) \in E(G)$. Hence, $N_{G_\varphi}(a) = \{b \mid \varphi(b) \in N_G(a)\} \setminus \{a\}$, and therefore:

- If $(a,\varphi(a)) \notin E(G)$, the map φ induces a bijection between $N_{G_\varphi}(a)$ and $N_G(a)$. In particular, $|N_{G_\varphi}(a)| = |N_G(a)|$.
- If $(a,\varphi(a)) \in E(G)$, the map φ induces a bijection between $N_{G_\varphi}(a)$ and $N_G \setminus \{\varphi(a)\}$. In particular, $|N_{G_\varphi}(a)| = |N_G(a)| - 1$.

In what follows, we take $\mathbb{G}_q$ to be the bipartite incidence graph of the projective plane $\mathrm{PG}(2,q)$ from the previous section. Let $\varphi : L(\mathbb{G}_q) \to R(\mathbb{G}_q)$ be defined as follows. Recall that a vertex $a \in L(\mathbb{G}_q)$ (that is, a point of the projective plane) corresponds to a 1-dimensional linear subspace of $\mathbb{F}_q^3$, while a vertex $\lambda \in R(\mathbb{G}_q)$ (that is, a line of the projective plane) corresponds to a 2-dimensional linear subspace of $\mathbb{F}_q^3$. Hence, for $a \in L(\mathbb{G}_q)$, we define $\varphi(a) \in R(\mathbb{G}_q)$ as the projective line corresponding to the 2-dimensional linear subspace orthogonal to a. Clearly, φ is bijective. It is also symmetric, since $a \in \varphi(b) \Leftrightarrow$ (a and b are orthogonal 1-dimensional linear subspaces) $\Leftrightarrow b \in \varphi(a)$. Note also that $(a,\varphi(a)) \in E(\mathbb{G}_q)$ if and only if a is self-orthogonal.

Let $\mathbb{G}_{q|\phi}$ be the bijective, symmetric reduction of $\mathbb{G}_q$ induced by φ. We will not use the explicit definition of φ, but only the fact it is bijective and symmetric. Note that $\mathbb{G}_{q|\phi}$ is a graph with q^2+q+1 vertices, and vertex degree

equal to either q (vertices corresponding to self-orthogonal linear subspaces) or $q+1$ (other vertices). The diameter of $\mathbb{G}_{q|\phi}$ is equal to 2, and for any two non-adjacent vertices $a, b \in V(\mathbb{G}_{q|\phi})$, there is a unique path of length 2 connecting them.

Theorem 4.4. Let k be such that $5k^2 - k - 10 \leqslant 2q$. Then $\mathbb{G}_{q|\phi}$ is k-vertex-minor universal.

Proof. We prove $\mathbb{G}_{q|\phi}$ satisfies the property (VMU$^\circ$) from Lemma 4.4. Consider a set of k vertices $K = \{u_1, u_2, \ldots, u_k\} \subseteq V(\mathbb{G}_{q|\phi})$. We will alternately refer to u_i's as points, since they are points of a projective plane (recall that $V(\mathbb{G}_{q|\phi}) = L(\mathbb{G}_q)$). To construct the sets $C = \{c_{ij} \mid 1 \leqslant i < j \leqslant k\} \subseteq V(\mathbb{G}_{q|\phi})$ and $S = \{a_{ij}, b_{ij} \mid 1 \leqslant i < j \leqslant k\} \subseteq V(\mathbb{G}_{q|\phi})$ from Lemma 4.4 we will proceed again by recursion, by running through pairs (u_i, u_j) in some particular order, say in lexicographical order with respect to indexes (i, j).

Fix some indexes (i, j), with $1 \leqslant i < j \leqslant k$, and assume that we have constructed (all neighbor sets below are defined with respect to the reduced graph $\mathbb{G}_{q|\phi}$):

(i) a set of vertices $C_{ij} = \{c_{i'j'} \mid (i', j') < (i, j)\} \subseteq V(\mathbb{G}_{q|\phi})$, such that C_{ij} is stable, $K \cap C_{ij} = \emptyset$, and $N_K(c_{i'j'}) = \emptyset$, for all $(i', j') < (i, j)$,

(ii) a set of vertices $S_{ij} = \{a_{i'j'}, b_{i'j'} \mid (i', j') < (i, j)\} \subseteq V(\mathbb{G}_{q|\phi})$, such that S_{ij} is stable, $S_{ij} \cap (K \cup C_{ij}) = \emptyset$, $N_{K \cup C}(a_{i'j'}) = \{u_{i'}, c_{i'j'}\}$ and $N_{K \cup C}(b_{i'j'}) = \{u_{j'}, c_{i'j'}\}$, for all $(i', j') < (i, j)$.

Note that at the first step of the recursion, i.e., for $(i, j) = (1, 2)$, we have $C_{1,2} = \emptyset$ and $S_{1,2} = \emptyset$. To proceed with the recursion, we need to construct c_{ij}, a_{ij}, and b_{ij}, which is done as follows (here below, for a set of points $P \subseteq L(\mathbb{G}_q)$, we denote by $\varphi(P) \subseteq R(\mathbb{G}_q)$ the image of P through φ).

- We choose $\alpha_{ij} \in R(\mathbb{G}_q) \setminus (\varphi(K) \cup \varphi(C_{ij}))$ a line passing through u_i, and not passing through any point in $(K \setminus \{u_i\}) \cup C_{ij} \cup S_{ij}$.
  There are $q+1$ lines passing through u_i, and to choose α_{ij} as above we need to avoid:
  - k lines in $\varphi(K)$ and $|C_{ij}| \leqslant k(k-1)/2 - 1$ lines in $\varphi(C_{ij})$,
  - at most $(k-1)$ lines passing through u_i and some point in $K \setminus \{u_i\}$,
  - at most $|C_{ij}|$ lines passing through u_i and some point in C_{ij},
  - at most $|S_{ij}| \leqslant k(k-1) - 2$ lines passing through u_i and some point in S_{ij}.

  In total, the number of lines to avoid is upper bounded by $|K| + |K \setminus \{u_i\}| + 2|C_{ij}| + |S_{ij}| \leqslant (2k-1) + 2(k(k-1) - 2) = 2k^2 - 5$. Thus, such a choice of α_{ij} is possible as long as $2k^2 - 5 \leqslant q$, which is ensured by the upper bound on the k value from the lemma.

  Let $I_{ij}^{(K)}$ be the set of intersection points between α_{ij} and the lines in $\varphi(K)$, and $I_{ij}^{(C)}$ be the set of intersection points between α_{ij} and the lines in $\varphi(C_{ij})$. Since $\alpha_{ij} \notin \varphi(K) \cup \varphi(C_{ij})$, it follows that it intersects each line in $\varphi(K)$ or $\varphi(C_{ij})$ in one point. Thus, $|I_{ij}^{(K)}| \leqslant |K| = k$ and $|I_{ij}^{(C)}| \leqslant |C_{ij}|$.

- We choose $\beta_{ij} \in R(\mathbb{G}_q) \setminus \varphi(C_{ij})$ a line passing through u_j, and not passing through any point in $(K \setminus \{u_j\}) \cup C_{ij} \cup S_{ij} \cup I_{ij}^{(K)} \cup I_{ij}^{(C)} \cup \{\varphi^{-1}(\alpha_{ij})\}$. There are $q+1$ lines passing through u_j, and to choose β_{ij} as above we need to avoid at most $|K \setminus \{u_j\}| + 2|C_{ij}| + |S_{ij}| + |I_{ij}^{(K)}| + |I_{ij}^{(C)}| + 1 \leqslant (k-1) + 2(k(k-1)-2) + k + \frac{1}{2}k(k-1) = \frac{1}{2}(5k^2 - k - 10)$ lines. Such a choice is possible as long as $5k^2 - k - 10 \leqslant 2q$, which is exactly the upper bound on the value of k from the lemma.

- Finally, we take c_{ij} to be the intersection point between lines α_{ij} and β_{ij}, $a_{ij} = \varphi^{-1}(\alpha_{ij})$ and $b_{ij} = \varphi^{-1}(\beta_{ij})$.

To complete the recursion, we need to prove that the sets $\bar{C}_{ij} := C_{ij} \cup \{c_{ij}\}$ and $\bar{S}_{ij} := S_{ij} \cup \{\alpha_{ij}, \beta_{ij}\}$ satisfy the conditions (i) and (ii), above.

For (i), we need to prove:

- $c_{ij} \notin C_{ij} \cup K$. This follows from the fact that the line α_{ij} does not pass through any point in $C_{ij} \cup (K \setminus \{u_i\})$ (and clearly, $c_{ij} \neq u_i$, since β_{ij} does not pass through u_i).

- c_{ij} is not adjacent (in $\mathbb{G}_{q|\phi}$) to any $c_{i'j'} \in C_{ij}$. Assume that c_{ij} is adjacent to some $c_{i'j'} \in C_{ij}$. It follows that $c_{ij} \in \varphi(c_{i'j'})$, and thus $c_{ij} \in I_{ij}^{(C)}$ (as it belongs to both α_{ij} and $\varphi(c_{i'j'})$), contradicting the fact that β_{ij} does not pass through any point in $I_{ij}^{(C)}$.

- $N_K(c_{ij}) = \emptyset$. Assume that c_{ij} is adjacent to some $a \in K$. It follows that $c_{ij} \in \varphi(a)$, and thus $c_{ij} \in I_{ij}^{(K)}$ (as it belongs to both α_{ij} and $\varphi(a)$), contradicting the fact that β_{ij} does not pass through any point in $I_{ij}^{(K)}$.

For (ii), we need to prove:

- $c_{ij} \notin S_{ij} \cup \{a_{ij}, b_{ij}\}$. First, $c_{ij} \notin S_{ij}$, which follows from the fact that α_{ij} does not pass through any point in S_{ij}. Assume that $c_{ij} = a_{ij}$. Then $u_i \in \alpha_{ij} = \varphi(c_{ij})$, meaning that u_i and c_{ij} are adjacent in $\mathbb{G}_{q|\phi}$, which is impossible, since we already proved that $N_K(c_{ij}) = \emptyset$. Assuming $c_{ij} = b_{ij}$ leads to a similar contradiction.

- $a_{ij} \notin K \cup C_{ij}$ and $b_{ij} \notin K \cup C_{ij}$. Assuming $a_{ij} \in K$, we get $c_{ij} \in \alpha_{ij} = \varphi(a_{ij})$, meaning that a_{ij} and c_{ij} are adjacent in $\mathbb{G}_{q|\phi}$, and thus contradicting $N_K(c_{ij}) = \emptyset$. Assuming $a_{ij} \in C_{ij}$ implies $\alpha_{ij} = \varphi(a_{ij}) \in \varphi(C_{ij})$, contradicting the choice of α_{ij}. Similarly, $b_{ij} \notin K \cup C_{ij}$.

- a_{ij} is not adjacent (in $\mathbb{G}_{q|\phi}$) to any $a_{i'j'}$ or $b_{i'j'} \in S_{ij}$. Assume that a_{ij} and $a_{i'j'}$ are adjacent. Then $a_{i'j'} \in \varphi(a_{ij}) = \alpha_{ij}$, contradicting the fact that α_{ij} does not pass through any point of S_{ij}. Assuming a_{ij} is adjacent to $b_{i'j'}$ leads to a similar contradiction.

- b_{ij} is not adjacent (in $\mathbb{G}_{q|\phi}$) to any $a_{i'j'}$ or $b_{i'j'} \in S_{ij}$. As above.

- a_{ij} and b_{ij} are not adjacent (in $\mathbb{G}_{q|\phi}$). Assuming they are adjacent, we get $a_{ij} \in \varphi(b_{ij}) = \beta_{ij}$, contradicting the fact that β_{ij} does not pass through $\varphi^{-1}(\alpha_{ij}) = a_{ij}$.

- $N_{K\cup C}(a_{ij}) = \{u_i, c_{ij}\}$. Since $u_i, c_{ij} \in \alpha_{ij} = \varphi(a_{ij})$, we have $\{u_i, c_{ij}\} \subseteq N_{K\cup C}(a_{ij})$. Assume that a_{ij} is incident to some vertex $a \in K\setminus\{u_i\}$. Then $a \in \varphi(a_{ij}) = \alpha_{ij}$, which is impossible, since α_{ij} does not pass through any point in $K \setminus \{u_i\}$. Assume that a_{ij} is incident to some vertex $c_{i'j'} \in C_{ij}$. Then $c_{i'j'} \in \varphi(a_{ij}) = \alpha_{ij}$, contradicting the fact that α_{ij} does not pass through any point of C_{ij}.

- $N_{K\cup C}(b_{ij}) = \{u_j, c_{ij}\}$. As above.

We conclude that $\bar{C}_{ij} = C_{ij} \cup \{c_{ij}\}$ and $\bar{S}_{ij} = S_{ij} \cup \{a_{ij}, b_{ij}\}$ satisfy the conditions (i) and (ii), above. By recursion, we can construct sets $C := \bar{C}_{k-1,k}$ and $S := \bar{S}_{k-1,k}$ satisfying the property (VMU$^\circ$) from Lemma 4.4, and thus we conclude that $\mathbb{G}_{q|\phi}$ is k-vertex-minor universal. □

4.4 Comparison between bipartite and reduced graph constructions

We denote by $\lceil x \rceil_{\mathrm{p}}$ the smallest prime power greater than or equal to a real number $x > 1$. For a given $k > 1$, let $q_2 := \left\lceil \frac{7}{4}k^2 - 4 \right\rceil_{\mathrm{p}}$ and $q_1 := \left\lceil \frac{5}{2}k^2 - \frac{1}{2}k - 5 \right\rceil_{\mathrm{p}}$, given by the inequalities in Theorem 4.3 (bipartite graph) and Theorem 4.4 (reduced graph), respectively. It follows that $\mathbb{G}_{q_2}$ is a k-vertex-minor universal of order $n_2 = 2(q_2^2 + q_2 + 1) \sim \frac{49}{8}k^4$, while $\mathbb{G}_{q_1|\phi}$ is a k-vertex-minor universal of order $n_1 = q_1^2 + q_1 + 1 \sim \frac{25}{4}k^4$ (where $\sim$ indicates asymptotic equivalence, as k goes to infinity). Thus, asymptotically, the bipartite graph construction yields k-vertex-minor universal graphs of slightly lower order than the reduced graph construction. Another interesting property of the bipartite graph is that the corresponding graph state $|\mathbb{G}_{q_2}\rangle$ is equivalent, up to local Clifford unitaries, to a Calderbank-Shor-Steane (CSS) state [3, Section IV]. However, to construct a desired graph on k-vertices of the bipartite-graph $\mathbb{G}_{q_2}$, we need to follow Lemma 4.3, thus to construct the sets $C_1, C_2, S_1, S_2, \Omega$ therein, which is done by following the steps highlighted in bold in the proof of Theorem 4.3. Note that this directly translates into a LOCC protocol to induce a desired graph state on k qubits of the state $|\mathbb{G}_{q_2}\rangle$, using Proposition 2.2. For the reduced graph the corresponding protocol is simpler, as we only have to construct the sets C, S from Lemma 4.4, which is again done by following the steps highlighted in bold in the proof of Theorem 4.4.

5 Conclusion

We showed the existence of k-vertex-minor universal graphs of order quadratic in k, which attain the optimum. This implies the existence of k-vertex-minor universal and thus k-pairable graph states with a quadratic number of qubits. Then, our study of the incidence graph of a finite projective plane exhibited two families of k-vertex-minor universal graphs of linear order in k^4. These two families being, to our knowledge, the first k-stabilizer universal quantum states, and so k-pairable quantum states, that can be constructed on a polynomial number of qubits in k.
This leaves open some questions for future work.

- The logical next step is the explicit, deterministic construction of an infinite family of k-vertex-minor universal graphs whose order is cubic, or even quadratic in k, asymptotically matching the order of the k-vertex-minor universal graphs which can be constructed in a probabilistic, non-deterministic way (although with arbitrarily high probability).

- Our probabilistic construction for k-vertex-minor universal graphs is asymptotically optimal. The graph states corresponding to $2k$-vertex-minor universal graphs are also k-pairable: however the only known lower bound on the size of k-pairable states (where one party holds only one qubit) is quasi-linear [2]. Does there exist k-pairable states with a quasi-linear number of qubits?

- Even though $2k$-stabilizer universality is a stronger requirement than k-pairability, it is not clear whether there exist k-pairable states which are not $2k$-stabilizer universal. A similar question can be asked for graphs: it is not clear whether there exist k-pairable graphs on more than 2 vertices which are not $2k$-vertex-minor universal.

- Stabilizer universality of graph states, when restricted to LOCC protocols using only local Clifford operations, local destructive Pauli measurements, and classical communication, is fully characterized by the combinatorial properties of the associated graph. Does it extend to stabilizer universality with arbitrary LOCC protocols? Does the stabilizer universality of graph states translate to an augmented version of k-vertex-minor universality that allows vertex disconnection? An analog question may be asked for pairability: is the underlying graph of a k-pairable graph state always k-pairable?

- We chose to focus on the ability to induce any stabilizer state on any subset of k qubit of a quantum state. What about the ability to induce any quantum state on any subset of k qubit? How does it relate to k-stabilizer universality?

- Bravyi et al. presented a construction of k-pairable states with an asymptotically optimal number of parties, in the case where each party holds at least 10 qubits [2]. How does k-stabilizer universality evolve when considering quantum communication networks where each party holds more than one qubit? Note that the construction of Bravyi et al. where each party holds at least 10 qubits does not translate well for k-stabilizer universality.

Acknowledgements

This work is supported by the PEPR integrated project EPiQ ANR-22-PETQ-0007 part of Plan France 2030, by the STIC-AmSud project Qapla' 21-STIC-10, by the QuantERA grant EQUIP ANR-22-QUA2-0005-01, and by the European projects NEASQC and HPCQS.

References

[1] Sara Bartolucci, Patrick Birchall, Hector Bombin, Hugo Cable, Chris Dawson, Mercedes Gimeno-Segovia, Eric Johnston, Konrad Kieling, Naomi Nickerson, Mihir Pant, et al. Fusion-based quantum computation. Nature Communications, 14(1):912, 2023. arXiv:2101.09310, doi:10.1038/s41467-023-36493-1.

[2] Sergey Bravyi, Yash Sharma, Mario Szegedy, and Ronald de Wolf. Generating k EPR-pairs from an n-party resource state. Quantum Information Processing, 2023. arXiv:2211.06497.

[3] Kai Chen and Hoi-Kwong Lo. Multi-partite quantum cryptographic protocols with noisy GHZ states. Quantum Information and Computation, 7(8), Nov 2007. arXiv:quant-ph/0404133, doi:10.26421/QIC7.8-1.

[4] Matthias Christandl, Vladimir Lysikov, Vincent Steffan, Albert H Werner, and Freek Witteveen. The resource theory of tensor networks, 2023. arXiv:2307.07394.

[5] Nathan Claudet, Mehdi Mhalla, and Simon Perdrix. Small k-pairable states, 2023. arXiv:2309.09956.

[6] Patricia Contreras-Tejada, Carlos Palazuelos, and Julio I. de Vicente. Asymptotic survival of genuine multipartite entanglement in noisy quantum networks depends on the topology. Physical Review Letters, 128(22), 2022. arXiv:2106.04634, doi:10.1103/physrevlett.128.220501.

[7] Axel Dahlberg, Jonas Helsen, and Stephanie Wehner. How to transform graph states using single-qubit operations: computational complexity and algorithms. Quantum Science and Technology, 5(4):045016, Sep 2020. arXiv:1805.05306, doi:10.1088/2058-9565/aba763.

[8] Axel Dahlberg, Jonas Helsen, and Stephanie Wehner. Transforming graph states to Bell-pairs is NP-Complete. Quantum, 4:348, Oct 2020. arXiv:1907.08019, doi:10.22331/q-2020-10-22-348.

[9] Axel Dahlberg and Stephanie Wehner. Transforming graph states using single-qubit operations. Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences, 376(2123):20170325, 2018. arXiv:1805.05305, doi:10.1098/rsta.2017.0325.

[10] Maarten Van den Nest, Jeroen Dehaene, and Bart De Moor. Graphical description of the action of local clifford transformations on graph states. Physical Review A, 69(2), Feb 2004. arXiv:quant-ph/0308151, doi:10.1103/physreva.69.022316.

[11] Gang Du, Tao Shang, and Jian-wei Liu. Quantum coordinated multi-point communication based on entanglement swapping. Quantum Information Processing, 16, Mar 2017. doi:10.1007/s11128-017-1558-2.

[12] Alex Fischer and Don Towsley. Distributing graph states across quantum networks. In IEEE International Conference on Quantum Computing and Engineering (QCE), pages 324–333, 2021. arXiv:2009.10888, doi:10.1109/QCE52317.2021.00049.

[13] Sobhan Ghanbari, Jie Lin, Benjamin MacLellan, Luc Robichaud, Piotr Roztocki, and Hoi-Kwong Lo. Optimization of deterministic photonic graph state generation via local operations, 2024. arXiv:2401.00635.

[14] Daniel Gottesman. The heisenberg representation of quantum computers, 1998. arXiv:quant-ph/9807006.

[15] Frederik Hahn, Anna Pappa, and Jens Eisert. Quantum network routing and local complementation. npj Quantum Information, 5(1):1–7, 2019. arXiv:1805.04559, doi:10.1038/s41534-019-0191-6.

[16] Marc Hein, Wolfgang Dür, Jens Eisert, Robert Raussendorf, Maarten Van den Nest, and Hans J. Briegel. Entanglement in graph states and its applications, 2006. arXiv:quant-ph/0602096.

[17] Jessica Illiano, Michele Viscardi, Seid Koudia, Marcello Caleffi, and Angela Sara Cacciapuoti. Quantum internet: from medium access control to entanglement access control, 2022. arXiv:2205.11923.

[18] Jérôme Javelle, Mehdi Mhalla, and Simon Perdrix. New protocols and lower bounds for quantum secret sharing with graph states. In Conference on Quantum Computation, Communication, and Cryptography, pages 1–12. Springer, 2012. arXiv:1109.1487.

[19] Donggyu Kim and Sang-il Oum. Vertex-minors of graphs: A survey, Oct 2023. URL: https://dimag.ibs.re.kr/home/sangil/wp-content/uploads/sites/2/2023/10/2023vertexminors-survey-revised.pdf.

[20] Seok-Hyung Lee and Hyunseok Jeong. Graph-theoretical optimization of fusion-based graph state generation. Quantum, 7:1212, Dec 2023. arXiv:2304.11988, doi:10.22331/q-2023-12-20-1212.

[21] Chao-Yang Lu, Xiao-Qi Zhou, Otfried Gühne, Wei-Bo Gao, Jin Zhang, Zhen-Sheng Yuan, Alexander Goebel, Tao Yang, and Jian-Wei Pan. Experimental entanglement of six photons in graph states. Nature physics, 3(2):91–95, 2007.

[22] Damian Markham and Barry C. Sanders. Graph states for quantum secret sharing. Physical Review A, 78:042309, 2008. arXiv:0808.1532, doi:10.1103/PhysRevA.78.042309.

[23] Clément Meignant, Damian Markham, and Frédéric Grosshans. Distributing graph states over arbitrary quantum networks. Physical Review A, 100:052333, Nov 2019. arXiv:1811.05445, doi:10.1103/PhysRevA.100.052333.

[24] Jorge Miguel-Ramiro, Alexander Pirker, and Wolfgang Dür. Optimized quantum networks. Quantum, 7:919, Feb 2023. arXiv:2107.10275, doi:10.22331/q-2023-02-09-919.

[25] Mihir Pant, Hari Krovi, Don Towsley, Leandros Tassiulas, Liang Jiang, Prithwish Basu, Dirk Englund, and Saikat Guha. Routing entanglement in the quantum internet. npj Quantum Information, 5(1):1–9, 2019. arXiv:1708.07142, doi:10.1038/s41534-019-0139-x.

[26] Eddie Schoute, Laura Mancinska, Tanvirul Islam, Iordanis Kerenidis, and Stephanie Wehner. Shortcuts to quantum network routing, 2016. arXiv:1610.05238.

[27] Péter Vrana and Matthias Christandl. Entanglement distillation from Greenberger–Horne–Zeilinger shares. Communications in Mathematical Physics, 352:621–627, 2017. arXiv:1603.03964.

A Some data on the size of the existence constraints

By Lemma 3.2, given some $k \in \mathbb{N} \setminus \{0\}$, there exists a k-vertex-minor universal bipartite graph G with $|L(G)| \geqslant k$, $|R(G)| \geqslant 4\binom{k}{2} + 5$ if

$$\left(\frac{k}{2^{|L(G)|-k+1}} + e^{-\frac{\left(\frac{|R(G)|}{4} - \binom{k}{2}+1\right)^2}{\left(\frac{7(|R(G)|-k)}{4} - \binom{k}{2}+1\right)}} \right) \times \binom{|L(G)| + |R(G)|}{k} < 1$$

In Table 1 we provide values for which there exists a k-vertex-minor universal bipartite graph of this order, for some particular values of k. In Table 2 we provide values for which a randomly generated bipartite graph is k-vertex-minor universal with at least 99% probability. Experimentally, adding a small, constant number of vertices to the randomly generated bipartite graph, greatly increases the probability of it to be k-vertex-minor universal.

Table 1: Parameters for which some k-vertex-minor universal bipartite graph exists.

k	3	4	5	6	7	8	9	10	11	12	13	14	15
|V(G)|	36	57	83	113	147	184	226	272	322	377	434	497	563
$\|L(G)\|$	18	24	32	40	48	55	63	72	80	90	97	107	115
$\|R(G)\|$	18	33	51	73	99	129	163	200	242	287	337	390	448

k	20	25	30	35	40	50	60	70	80	90	100
|V(G)|	955	1448	2041	2736	3531	5424	7718	10414	13512	17012	20912
$\|L(G)\|$	161	208	256	306	357	461	568	677	788	902	1016
$\|R(G)\|$	794	1240	1785	2430	3174	4963	7150	9737	12724	16110	19896

B Complete proof of Theorem 4.3

Theorem 4.3. Let k be such that $7k^2 - 16 \leqslant 4q$. Then $\mathbb{G}_q$ is k-vertex-minor universal.

Proof. We want to prove that $\mathbb{G}_q$ satisfies the property (VMU*) of Lemma 4.3. Let $K = \{u_1, \ldots, u_{k_1}, \lambda_1, \ldots, \lambda_{k_2}\}$ be a set of k points with $k_1 \leqslant k$, $\{u_1, \ldots, u_{k_1}\} \subseteq L(\mathbb{G}_q)$, and $\{\lambda_1, \ldots, \lambda_{k_2}\} \subseteq R(\mathbb{G}_q)$. The proof is made of 3 inductions to construct: (1) the subsets $C_1 \subseteq L(\mathbb{G}_q)$ and $S_1 \subseteq R(\mathbb{G}_q)$, (2) the subset $\Omega \subseteq R(\mathbb{G}_q)$, and finally (3) the subsets $C_2 \subseteq R(\mathbb{G}_q)$ and $S_2 \subseteq L(\mathbb{G}_q)$, all defined as in the property (VMU*) from Lemma 4.3.

Table 2: Parameters for which a randomly generated bipartite graph is k-vertex-minor universal with at least 99% probability.

k	3	4	5	6	7	8	9	10	11	12	13	14	15
|V(G)|	47	68	93	123	156	194	235	281	331	385	443	505	571
$\|L(G)\|$	25	32	39	47	55	63	71	79	88	96	105	113	122
$\|R(G)\|$	22	36	54	76	101	131	164	202	243	289	338	392	449

k	20	25	30	35	40	50	60	70	80	90	100
|V(G)|	962	1456	2049	2743	3539	5431	7726	10422	13519	17019	20920
$\|L(G)\|$	167	215	263	313	364	468	575	684	795	908	1023
$\|R(G)\|$	795	1241	1786	2430	3175	4963	7151	9738	12724	16111	19897

(1) We construct the subsets $C_1 = \{c_{ij} \mid 1 \leqslant i < j \leqslant k_1\} \subseteq L(\mathbb{G}_q)$ and $S_1 = \{\alpha_{ij}, \beta_{ij} \mid 1 \leqslant i < j \leqslant k_1\} \subseteq R(\mathbb{G}_q)$. The first induction is done on pairs (u_i, u_j) in the lexicographical order over the indices and with the next induction hypothesis:

(IH_1^{ij}) The subset of points $C_1^{ij} := \{c_{i'j'} \mid (i', j') < (i, j)\} \subseteq L(\mathbb{G}_q)$ and the subset of lines $S_1^{ij} = \{\alpha_{i'j'}, \beta_{i'j'} \mid (i', j') < (i, j)\} \subseteq R(\mathbb{G}_q)$ are such that

(i) $C_1^{ij} \cap K_1 = \emptyset$,

(ii) $S_1^{ij} \cap K_2 = \emptyset$,

(iii) $N_{K_2}(C_1^{ij}) = \emptyset$,

(iv) $N_{K_1 \sqcup C_1^{ij}}(\alpha_{i'j'}) = \{u_{i'}, c_{i'j'}\}, N_{K_1 \sqcup C_1^{ij}}(\beta_{i'j'}) = \{u_{j'}, c_{i'j'}\}$, for all $(i', j') < (i, j)$.

The initialisation is done for $i = 1$ and $j = 2$. The subsets $S_1^{12} = \emptyset$ and $C_1^{12} = \emptyset$ satisfy the induction hypotheses (IH_1^{12}).

Then, for each induction step (i, j) with $1 \leqslant i < j \leqslant k$. We construct α_{ij}, β_{ij} and c_{ij}, as follows.

- We choose a line α_{ij} that does not belong to K_2 and passes only through u_i in $C_1^{ij} \sqcup K_1$. There are $q + 1$ lines passing through u_i (the degree d° of a vertex is constant above $\mathbb{G}_q$), at most k_2 of them belong to K_2. Moreover, the points of $C_1^{ij} \sqcup K_1 \setminus \{u_i\}$, paired with u_i, define at most $k_1(k_1 - 1)/2 - 1 + k_1 - 1$ lines to avoid. Such line exists if $\underbrace{q+1}_{d^\circ} - \underbrace{k_2}_{|N_{K_2}(u_i)| \leqslant} \geqslant \underbrace{1}_{\alpha_{ij}} + \underbrace{k_1 - 1}_{|K_1 \setminus \{u_i\}|} + \underbrace{k_1(k_1-1)/2 - 1}_{|C_1^{ij}| \leqslant}$ so if $k_1^2 + 2k - k_1 - 4 \leqslant 2q$. Clearly, the line α_{ij} does not belong to S_1^{ij} otherwise it would pass through one point of C_1^{ij}. Then the choice of β_{ij} is driven by c_{ij}, its intersection point with α_{ij}. Indeed, the point c_{ij} can not belong to a line of K_2, to a line of S_1^{ij}, and to the subset C_1^{ij}, the two first constraints are summarized by avoiding the two next sets: Let $I_{S_1}^{ij} \subseteq L(\mathbb{G}_q)$ be the set of intersection points of α_{ij} with lines of S_1^{ij}, with cardinality $|I_{S_1}^{ij}| \leqslant |S_1^{ij}| \leqslant k_1(k_1 - 1) - 2$. Let $I_{K_2}^{ij} \subseteq L(\mathbb{G}_q)$ be the set of intersection points of α_{ij} with lines of K_2, with cardinality $|I_{K_2}^{ij}| \leqslant |K_2| \leqslant k_2$.

- Thus, we choose a line β_{ij} that does not belong to K_2 and passes only through u_j in $K_1 \sqcup C_1^{ij} \sqcup I_{S_1}^{ij} \sqcup I_{K_2}^{ij}$. Such choice is possible if

$$\underbrace{q+1}_{d^\circ} - \underbrace{k_2}_{|N_{K_2}(u_j)|\leqslant} \geqslant \underbrace{1}_{\beta_{ij}} + \underbrace{k_1-1}_{|K_1\setminus\{u_j\}|} + \underbrace{k_1(k_1-1)/2-1}_{|C_1^{ij}|\leqslant} + \underbrace{k_1(k_1-1)-2}_{|I_{S_1}^{ij}|\leqslant} + \underbrace{k_2}_{|I_{K_2}^{ij}|\leqslant}$$

  and so if $2q \geqslant 3k_1^2+4k-5k_1-8$. The line β_{ij} does not belong to $S_1^{ij} \sqcup \{\alpha_{ij}\}$ otherwise it would pass through a point of C_1^{ij}.

- Finally, we take the point c_{ij} to be intersection point of α_{ij} and β_{ij}. The point c_{ij} does not belong to C_1^{ij} as neither one of α_{ij} nor β_{ij} passes through the points in C_j.

To end the first recursion, we need to prove:

(i) $c_{ij} \notin C_1^{ij} \sqcup K_1$. It follows from the line α_{ij} not passing through any point in $C_1^{ij} \sqcup K_1 \setminus \{u_i\}$ and the line β_{ij} not passing through u_i.

(ii) $\alpha_{ij}, \beta_{ij} \notin S_1^{ij} \sqcup K_2$. It follows from the avoidance of the points of C_1^{ij} during the choice of the lines α_{ij} and β_{ij}.

(iii) $N_{K_2}(c_{ij}) = \emptyset$. The point c_{ij} does not belong to a line of K_2 as points of $I_{K_2}^{ij}$ are avoided when β_{ij} is chosen.

(iv) $N_{K_1 \sqcup (C_1^{ij} \sqcup \{c_{ij}\})}(\alpha_{i'j'}) = \{u_{i'}, c_{i'j'}\}$, $N_{K_1 \sqcup (C_1^{ij} \sqcup \{c_{ij}\})}(\beta_{i'j'}) = \{u_{j'}, c_{i'j'}\}$, $\forall (i',j') \leqslant (i,j)$. First, the line α_{ij} (resp. β_{ij}) passes only through the points u_i (resp. u_j) and c_{ij} in the set $K_1 \sqcup C_1^{ij}$. Then, for $(i',j') < (i,j)$, the lines $\alpha_{i'j'}$ and $\beta_{i'j'}$ do not pass through the point c_{ij} otherwise c_{ij} would belong to the subset $I_{S_1}^{ij}$ which is avoided during the choice of the line β_{ij}.

By recursion, we constructed the subsets $C_1 := C_1^{k_1,k_1}$ and $S_1 := S_1^{k_1,k_1}$ satisfying the first two constraints of the property (VMU*) from Lemma 4.3.

(2) Next step, we construct the subset $\Omega = \{\omega_{ij} \mid 1 \leqslant i \leqslant k_1, 1 \leqslant j \leqslant k_2\}$. We want one vertex ω_{ij} per pair (u_i, λ_j) between K_1 and K_2. This set is constructed inductively for each value of $1 \leqslant i \leqslant k_1$.

(IH$_2^i$) The subset of lines $\Omega^i = \{\omega_{i'j}, i' < i, 1 \leqslant j \leqslant k_2\} \subseteq R(\mathbb{G}_q)$ is such that

(v) $(K_2 \sqcup S_1) \cap \Omega^i = \emptyset$

(vi) $N_{K_1 \sqcup C_1}(\omega_{i'j}) = \{u_{i'}\}$ for all $1 \leqslant i' < i$ and $1 \leqslant j \leqslant k_2$.

The initialisation is done for $i = 1$, $\Omega^0 = \emptyset$ satisfies (IH$_2^i$).

Then, for $1 \leqslant i \leqslant k_1$. We choose k_2 distinct vertices $\{\omega_{ij}\}_{1 \leqslant j \leqslant k_2}$ adjacent to the vertex u_i (one for each vertex λ_j of K_2) such that $\omega_{ij} \notin K_2 \sqcup S_1 \sqcup \Omega^i$ and $N_{K_1 \sqcup C_1}(\omega_{ij}) = \{u_i\}$. The vertices of S_1 don't add constraints in the equation as they are already taken into account by the avoidance of vertices in C_1. Thus, such choice is possible if

$$\underbrace{q+1}_{d^\circ} - \underbrace{k_2}_{|N_{K_2}(u_i)|\leqslant} - \underbrace{0}_{|N_{\Omega^i}(u_i)|} \geqslant \underbrace{k_2}_{\forall j, \omega_{ij}} + \underbrace{k_1-1}_{|K_1\setminus\{u_i\}|} + \underbrace{k_1(k_1-1)/2}_{|C_1|}$$

and so if $2q \geqslant k_1^2 + 4k - 3k_1 - 4$ (where k_2 has been replaced by $k - k_1$). By construction, ω_{ij} avoid all points $u_{i'}$ for $i' \neq i$ and thus is distinct of $\omega_{i'j'}$ for $1 \leqslant i' < i$ and $1 \leqslant jj' \leqslant k_2$.

To end the induction, we need to prove:

(v) $\{\omega_{ij}, 1 \leqslant j \leqslant k_2\} \cap (\Omega^i \sqcup K_2 \sqcup S_1) = \emptyset$. By construction.

(vi) $N_{K_1 \sqcup C_1}(\omega_{i'j}) = \{u_{i'}\}$ for all $1 \leqslant i' \leqslant i$ and $1 \leqslant j \leqslant k_2$. First, by construction the vertex ω_{ij} has only the vertex u_i as neighbor in $K_1 \sqcup C_1$. For $i' < i$, nothing has changed.

By induction, we have constructed the set $\Omega := \Omega^{k_1+1}$ that satisfies the third constraint of (VMU*) from Lemma 4.3.

The remaining part of the proof consists in doing a truncated right k-vertex-minor universality with constraints coming from the already defined subsets K_1, K_2, S_1, C_1, and Ω. It will be really similar to the first part of this proof up to a swapping of the role of points and lines, swapping which is possible because of the symmetry property of the incidence graph of the finite projective plane.

(3) Finally, we build the sets $C_2 = \{\gamma_{i,j}, 1 \leqslant j \leqslant k_2 \mid j < i \leqslant k_1 + k_2\} \subseteq R(\mathbb{G}_q)$ of lines and $S_2 = \{a_{ij}, b_{ij}, 1 \leqslant j \leqslant k_2, j < i \leqslant k_1 + k_2\} \subseteq L(\mathbb{G}_q)$ of points. Indeed, we only need to control a subset of the edges in $K_2 \sqcup \Omega$, the subset of edges with two ends in K_2 and the subset of edges $\{(\omega_{ij}, \lambda_j)\}_{1 \leqslant i \leqslant k_1, 1 \leqslant j \leqslant k_2}$ with one end in K_2 and the other in Ω. Thus, the induction is done according to the lexicographical order on the indexes (i, j) for $1 \leqslant j \leqslant k_2$ and $j < i \leqslant k_2 + k_1$ where (i, j) is used to index the pair (λ_i, λ_j) if $i \leqslant k_2$, otherwise $(\omega_{(i-k_2)j}, \lambda_j)$. In the induction, we denote the by r_{ij} the vertex of $R(\mathbb{G}_q)$ that corresponds to λ_i if $i \leqslant k_2$ otherwise $\omega_{(i-k_2)j}$. The induction hypothesis is:

(IH$_3^{ij}$) The set of lines $C_2^{ij} = \{\gamma_{i'j'} \mid (i', j') < (i, j)\} \subseteq R(\mathbb{G}_q)$ and the set of points $S_2^{ij} = \{a_{i'j'}, b_{i'j'} \mid (i', j') < (i, j)\} \subseteq L(\mathbb{G}_q)$ are such that

(vii) $(K_2 \sqcup S_1 \sqcup \Omega) \cap C_2^{ij} = \emptyset$,

(viii) $N_{C_2^{ij}}(K_1 \sqcup C_1) = \emptyset$,

(ix) $(K_1 \sqcup C_1) \cap S_2^{ij} = \emptyset$.

(x) $N_{K_2 \sqcup S_1 \sqcup \Omega \sqcup C_2^{ij}}(a_{i'j'}) = \{r_{i'j'}, \gamma_{i'j'}\}$, $\quad N_{K_2 \sqcup S_1 \sqcup \Omega \sqcup C_2^{ij}}(b_{i'j'}) = \{\lambda_{j'}, \gamma_{i'j'}\}$ for $(i', j') < (i', j')$.

The initialisation is done for $i = 2$ and $j = 1$. We have that $C_2^{ij} = \emptyset$ and $S_2^{ij} = \emptyset$ satisfy (IH$_3^{21}$). We then construct a_{ij}, b_{ij}, and γ_{ij}. The steps are the same as for the first induction so less detailed.

- We choose a point a_{ij} not in $N_{K_1 \sqcup C_1}(r_{ij})$ and that does not belong to any other line than r_{ij} in $K_2 \sqcup S_1 \sqcup \Omega \sqcup C_2^{ij}$. Such point exists if

$$\underbrace{q+1}_{d^\circ} - \underbrace{k_1}_{|N_{K_1}(r_{ij})| \leqslant} - \underbrace{0}_{|N_{C_1}(r_{ij})|} \geqslant \underbrace{1}_{a_{ij}} + \underbrace{k_2 + k_1 k_2 - 1}_{|(K_2 \sqcup \Omega) \setminus \{r_{ij}\}|} + \underbrace{k_1(k_1 - 1)}_{|S_1|} + \underbrace{k_2(k_2 - 1)/2 + k_1 k_2 - 1}_{|C_2^{ij}|}$$

  and thus if $2q \geqslant k^2 - k_1^2 + 2kk_1 + k - k_1 - 4$. With similar arguments on the double role of C_2^{ij}, $a_{ij} \notin S_2^{ij} \sqcup K_1 \sqcup C_1$. Then, the choice of b_{ij} is driven by

some constraints on the line passing through a_{ij} and b_{ij}. Indeed, the line γ_{ij} can not belong to the subset $I^{ij}_{S_2} \subseteq R(\mathbb{G}_q)$ of lines defined by a_{ij} and each point of S^{ij}_2 and of cardinality $|I^{ij}_{S_2}| \leqslant k_2(k_2-1)+2k_1k_2-2$. The line γ_{ij} can not also belong to the subset $I^{ij}_{K_1\sqcup C_1} \subseteq R(\mathbb{G}_q)$ of lines defined by a_{ij} with each point of $K_1 \sqcup C_1$ and of cardinality $|I^{ij}_{K_1\sqcup C_1}| \leqslant k_1 + k_1(k_1-1)/2$.

- Then, we choose the point b_{ij} that does not belong to $N_{K_1\sqcup C_1}(\lambda_j)$ and that does not belong to any other line that λ_j in $K_2 \sqcup S_1 \sqcup \Omega \sqcup C^{ij}_2 \sqcup I^{ij}_{S_2} \sqcup I^{ij}_{K_1\sqcup C_1}$. Such choice of point is possible if

$$\underbrace{q+1}_{d^\circ} - \underbrace{k_1}_{|N_{K_1}(\lambda_j)|\leqslant} - \underbrace{0}_{|N_{C_1}(\lambda_j)|} \geqslant \underbrace{1}_{b_{ij}} + \underbrace{k_2-1}_{|K_2\setminus\{\lambda_j\}|} + \underbrace{k_1k_2}_{|\Omega|} + \underbrace{k_1(k_1-1)}_{|S_1|}$$
$$+ \underbrace{k_2(k_2-1)/2 + k_1k_2 - 1}_{|C^{ij}_2|} + \underbrace{k_2(k_2-1)+2k_1k_2-2}_{|I^{ij}_{S_2}|\leqslant} + \underbrace{k_1 + k_1(k_1-1)/2}_{|I^{ij}_{K_1\sqcup C_1}|\leqslant}$$

  and so if $2q \geqslant 3k^2 - 2k_1^2 + 2kk_1 + 2k_1 - k - 8$. With similar arguments as before, b_{ij} does not belong to $S^{ij}_2 \sqcup \{a_{ij}\} \sqcup K_1 \sqcup C_1$.

- Finally, we take γ_{ij} to be the line passing through a_{ij} and b_{ij}. It is then clear that, γ_{ij} does not belong to C^{ij}_2.

To end this last induction, we need to prove:

(vii) $\gamma_{ij} \notin (C^{ij}_2 \sqcup K_2 \sqcup S_1 \sqcup \Omega) = \emptyset$. It follows from the construction as the point a_{ij} (resp. b_{ij}) does not belong to any other line that r_{ij} (resp. λ_j) in $K_2 \sqcup S_1 \sqcup \Omega$ and r_{ij} is always distinct from λ_j.

(viii) $N_{K_1\sqcup C_1}(\gamma_{ij}) = \emptyset$. It follows from $\gamma_{ij} \notin I^{ij}_{K_1\sqcup C_1}$ as the point b_{ij} can not belong to lines of $I^{ij}_{K_1\sqcup C_1}$.

(ix) $a_{ij}, b_{ij} \notin (S^{ij}_2 \sqcup K_1 \sqcup C_1)$. It is implied by construction constraints.

(x) $N_{K_2\sqcup S_1\sqcup\Omega\sqcup C^{ij}_2\sqcup\gamma_{ij}}(a_{i'j'}) = \{r_{i'j'}, \gamma_{i'j'}\}$, $\quad N_{K_2\sqcup S_1\sqcup\Omega\sqcup C^{ij}_2\sqcup\gamma_{ij}}(b_{i'j'}) = \{\lambda_{j'}, \gamma_{i'j'}\}$ for $(i',j') < (i',j')$. First, by construction a_{ij} (resp. b_{ij}) does not belong to any other lines that r_{ij} (resp. λ_j) and γ_{ij} in $K_2 \sqcup \Omega \sqcup S_1 \sqcup C_2$. For $(i',j') < (i,j)$, the points $a_{i'j'}$ and $b_{i'j'}$ do not belong to the lines γ_{ij} otherwise γ_{ij} would belong to the subset $I^{ij}_{S_2}$ which is avoided during the choice of the point b_{ij}.

By induction, we have constructed $S_2 := S_2^{(k_2+k_1)k_2} + \{a_{(k_2+k_1)k_2}, b_{(k_2+k_1)k_2}\}$ and $C_2 := C_2^{(k_2+k_1)k_2} + \{\gamma_{(k_2+k_1)k_2}\}$ satisfying the last two constraints of (VMU*) from Lemma 4.3. Thus, if all bounds on q are satisfied then all constraints from (VMU*) are satisfied and then Lemma 4.3 directly gives us that $\mathbb{G}_q$ is k-vertex-minor universal. As Theorem 4.3 is stated only with k, the bounds on q have to be respected for all value of k_1 between 0 and $k/2$ (or between $k/2$ and k). To get the constraints independent of k_1, some basic calculus give us:

- $2q \geqslant k^2/4 + 3k/2 - 4$ for α_{ij},
- $2q \geqslant 3k^2/4 + 3k/2 - 8$ for β_{ij},

- $2q \geqslant k^2/4 + 5k/2 - 4$ for ω_{ij},
- $2q \geqslant 9k^2/4 + k/2 - 4$ for a_{ij},
- $2q \geqslant 7k^2/2 - 8$ for b_{ij}.

Finally, for $k \geqslant 2$, the strongest one is $2q \geqslant 7k^2/2 - 8$. So, $\mathbb{G}_q$ is k-vertex-minor universal if $4q \geqslant 7k^2 - 16$. □